

Genomics data analysis

基因组学数据分析

线性回归、分布图

李程

北京大学生命科学学院、
生物信息中心、统计科学中心

邮件: cheng_li@pku.edu.cn

主页: <http://3d-genome.life>



课程网页

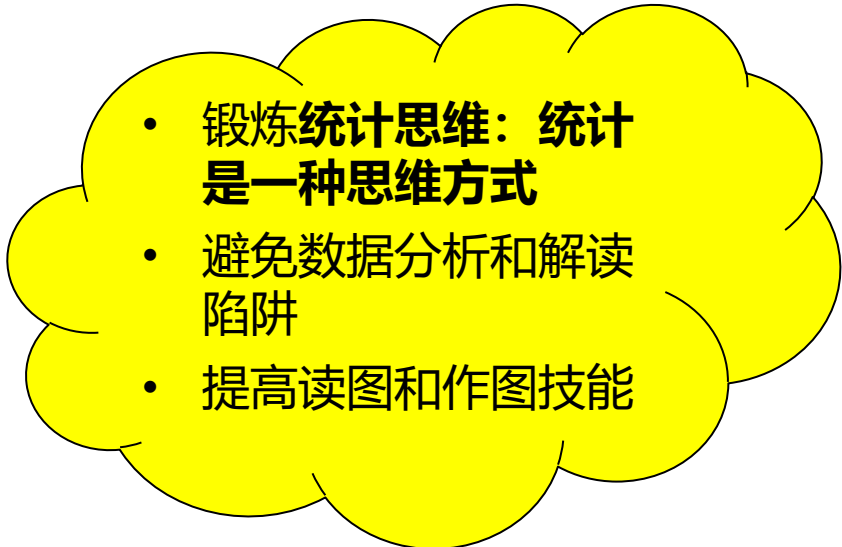
2024年秋季助教:

待定

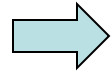
周二下午3-5点, 北大理教306 | 联系教师加入课程微信群

目标问题

1. 统计思维从什么角度看待数据？
2. 一个变量的差异性可以被另一个变量解释的比例 (proportion of variation explained) 指什么？
3. 统计作图（如散点图）的基本要素是什么？

- 
- 锻炼统计思维：统计是一种思维方式
 - 避免数据分析和解读陷阱
 - 提高读图和作图技能

本节内容



- 线性回归举例
- 线性回归**模型**
- 统计**陷阱**：均值回归效应
- 统计**作图**：数据分布
- 助教演示（课后练习）：
 - RStudio、文件读取、作图

Correlation between Tumor Mutational Burden and Objective Response Rate

组学思路:

- 整合不同公共数据集, 发现关联, 做预测或提示机制
- 散点图表示四维数据

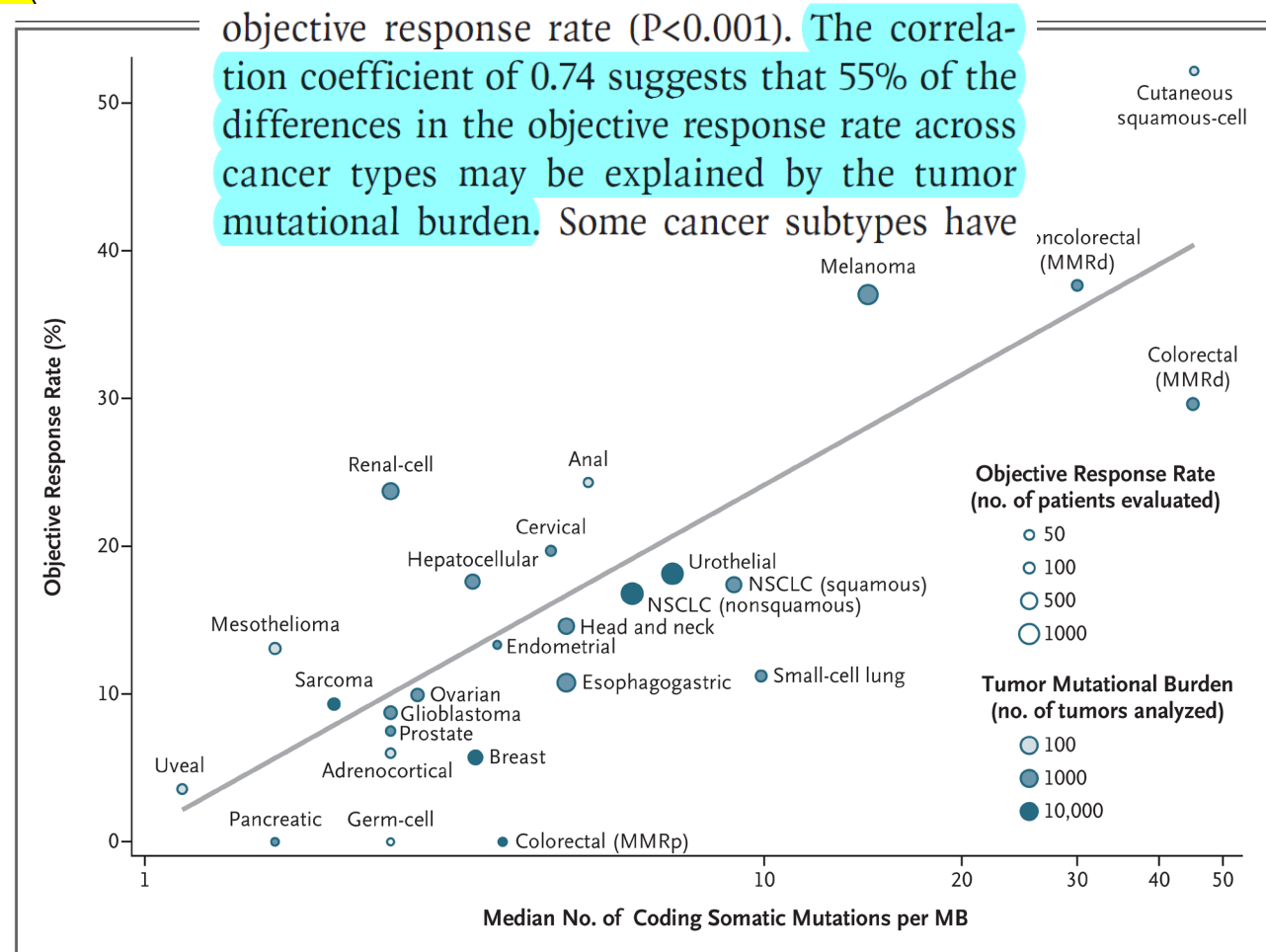


Figure 1. Correlation between Tumor Mutational Burden and Objective Response Rate with Anti-PD-1 or Anti-PD-L1 Therapy in 27 Tumor Types.

Shown is the median number of coding somatic mutations per megabase (MB) of DNA in 27 tumor types or subtypes among patients who received inhibitors of programmed death 1 (PD-1) protein or its ligand (PD-L1), as described in published studies for which data regarding the objective response rate are available. The number of pa-

癌症病因的“坏运气”学说

CANCER ETIOLOGY

Variation in cancer risk among tissues can be explained by the number of stem cell divisions



典型统计模型
的结论

Cristian Tomasetti^{1*} and Bert Vogelstein^{2*}

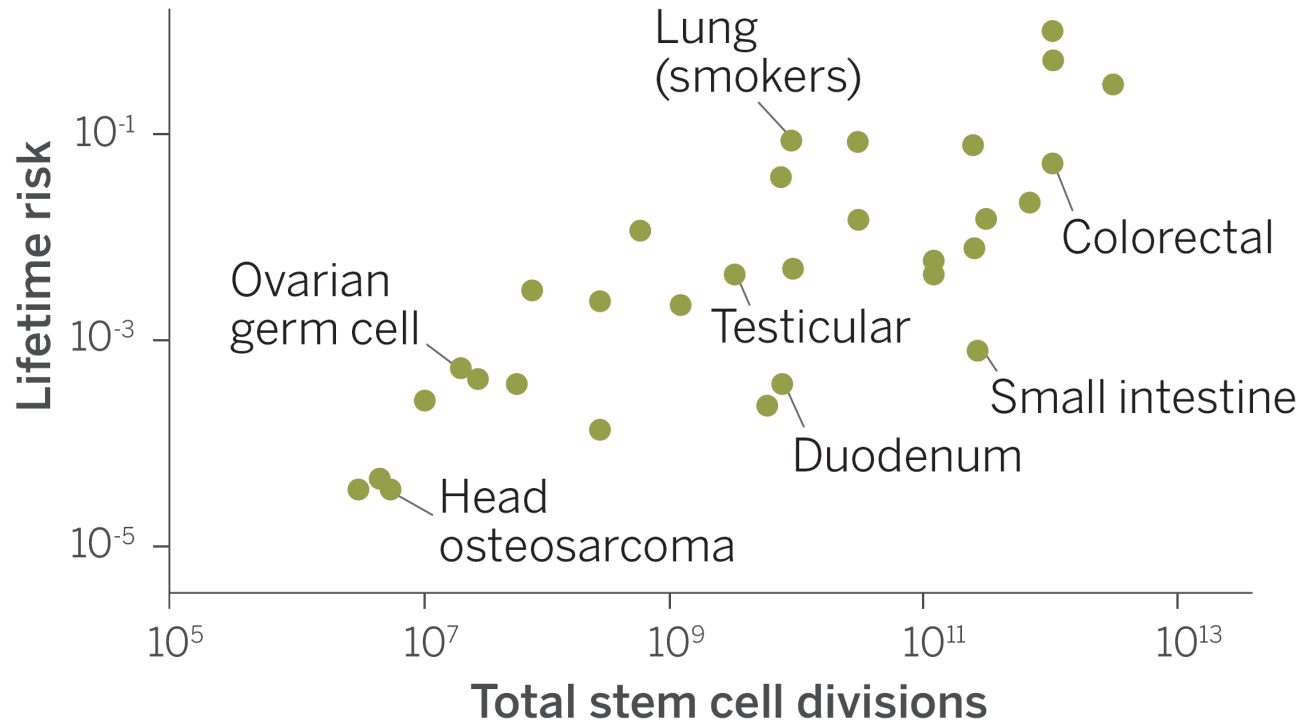
Some tissue types give rise to human cancers millions of times more often than other tissue types. Although this has been recognized for more than a century, it has never been explained. Here, we show that the lifetime risk of cancers of many different types is strongly correlated (0.81) with the total number of divisions of the normal self-renewing cells maintaining that tissue's homeostasis. These results suggest that only a third of the variation in cancer risk among tissues is attributable to environmental factors or inherited predispositions. The majority is due to “bad luck,” that is, random mutations arising during DNA replication in normal, noncancerous stem cells. This is important not only for understanding the disease but also for designing strategies to limit the mortality it causes.

Science, 2015, Vol 347, p.78

癌症发生率和干细胞分裂

Charting cancer risk

As the number of stem cell divisions in a tissue rises, so does the chance of cancer striking that site.



“2/3的癌症差异性由于**“坏运气”**，由正常成体干细胞的DNA复制中产生的随机突变造成”（1/3由于环境和遗传因素）

Science, 2015, Vol 347, p.78

CANCER ETIOLOGY

论文试图区分遗传、环境、运气（干细胞复制）对癌症发生的贡献大小

Stem cell divisions, somatic mutations, cancer etiology, and cancer prevention

Cristian Tomasetti,^{1,2*} Lu Li,² Bert Vogelstein^{3*}

Cancers are caused by mutations that may be inherited, induced by environmental factors, or result from DNA replication errors (R). We studied the relationship between the number of normal stem cell divisions and the risk of 17 cancer types in 69 countries throughout the world. The data revealed a strong correlation (median = 0.80) between cancer incidence and normal stem cell divisions in all countries, regardless of their environment. The major role of R mutations in cancer etiology was supported by an independent approach, based solely on cancer genome sequencing and epidemiological data, which suggested that R mutations are responsible for two-thirds of the mutations in human cancers. All of these results are consistent with epidemiological estimates of the fraction of cancers that can be prevented by changes in the environment. Moreover, they accentuate the importance of early detection and intervention to reduce deaths from the many cancers arising from unavoidable R mutations.

Science, 2017, Vol 355, p.1330

新闻报道

NEWS | IN DEPTH

BIOMEDICINE

The bad luck of cancer

Analysis suggests most cases can't be prevented

By Jennifer Couzin-Frankel

研究说的是66%的癌症相关基因突变是基因复制中的随机差错导致的，不能等同于“66%的癌症就是因为患者运气差造成的”。



Tomasetti: “我们不是评估随机突变导致的癌症的比例，而是计算致癌突变中随机突变的比例。”

论文争议

- 引发多篇讨论文章
 - 生物学假设、使用的数据、统计方法
 - 相关性vs.因果关系
 - 多个基因突变导致癌症
 - **作者、媒体如何避免误导**，如15年论文：65%并不是“所有癌症发病的65%”，而是“人体不同器官癌症发生率差异中的65%”

Comment in

Biomedicine. The bad luck of cancer. [Science. 2015]

Cancer: Risk factors and random chances. [Nature. 2015]

Cancer risk: role of chance overstated. [Science. 2015]

Cancer risk: tumors excluded. [Science. 2015]

Cancer risk: prevention is crucial. [Science. 2015]

Do mutational dynamics in stem cells explain the origin of common cancers? [Cell Stem Cell. 2015]

Cancer risk: role of environment. [Science. 2015]

Cancer risk: many factors contribute. [Science. 2015]

Cancer risk: accuracy of literature. [Science. 2015]

Cancer risk: role of environment—response. [Science. 2015]

Cancer: mixed messages, common purpose. [Lancet. 2015]

[Cancer—not just bad luck!]. [Tidsskr Nor Lægeforen. 2015]

Is 'bad luck' an important determinant of cancer incidence and does this concept apply to kidney tumors

Bad luck and cancer: Does evolution spin the wheel of fortune? [Bioessays. 2015]

The contribution of lifestyle, environment, genetics and chance to cancer risk in individuals and populati

Cancer Etiology, Bad Luck, and Ecological Fallacy. [World Neurosurg. 2015]

[Stem cell renewal and cancer epidemiology – about a paper published in Science by Tomasetti and Vo

[A New Year scientific meteor]. [Epidemiol Prev. 2015]

Somatic mutations may not be the primary drivers of cancer formation. [Int J Cancer. 2015]

A Critical Examination of the "Bad Luck" Explanation of Cancer Risk. [Cancer Prev Res (Phila). 2015]

Stem cell divisions and cancer. [Leukemia. 2015]

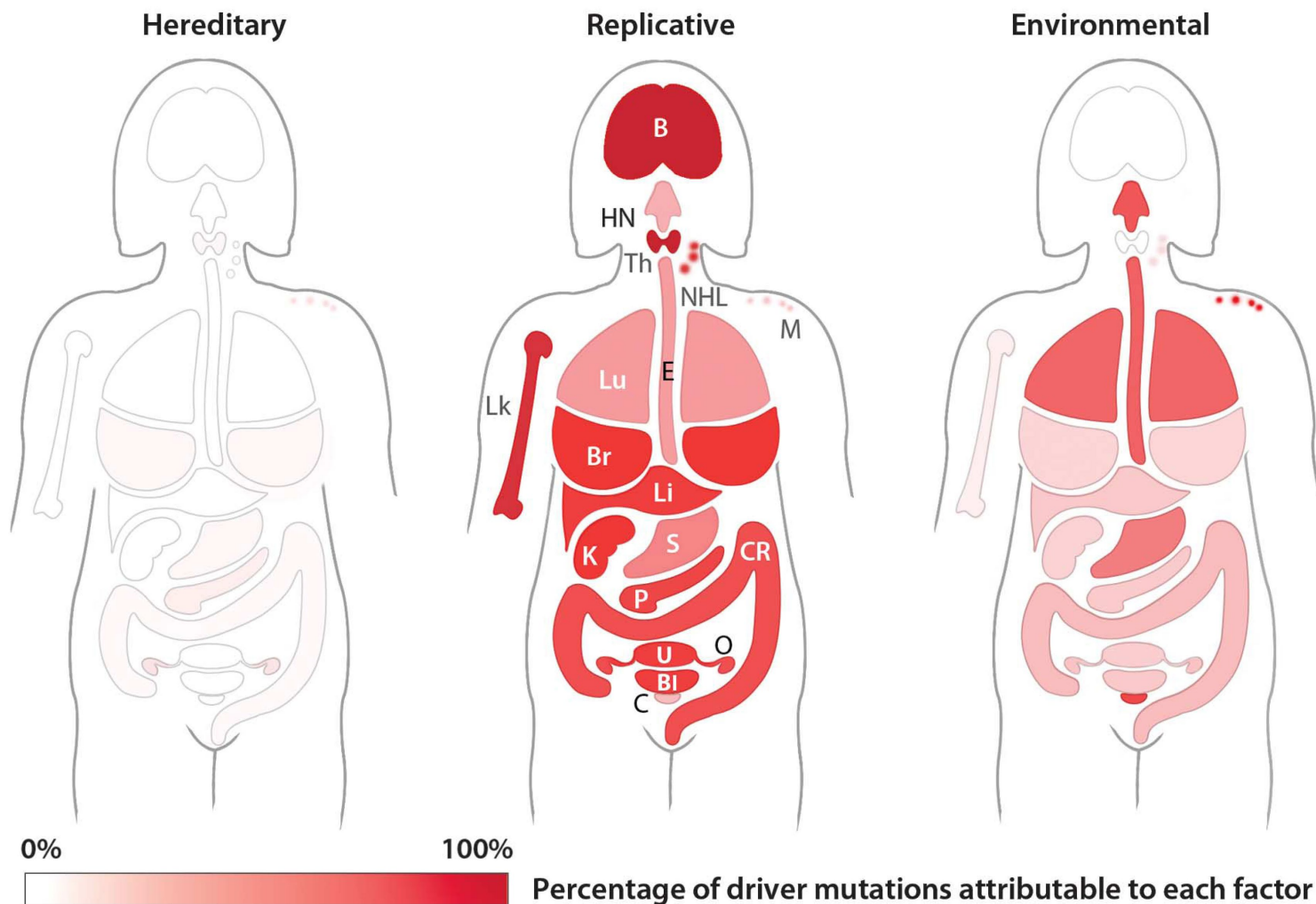
Are Most Cancers Caused by Specific Risk Factors Acting on Tissues With High Underlying Stem Cell I

Breast Cancer Prevention: Can Women's Expectations Be Met? [Oncologist. 2016]

Of Cancer and Random Mutations. [Cancer Discov. 2017]

<https://www.ncbi.nlm.nih.gov/pubmed/25554788>

<https://www.jianshu.com/p/5adfdfac0d5>



对于18种代表性癌症类型，示意图描述了遗传（左）、DNA复制（中）、环境因素（右）的影响比例。三个因素比例总和是100%，跨度为白色（0%）至最亮红色（100%）

癌症到底能不能预防？

癌症类型	环境	遗传	复制（随机）
宫颈	74.6%	0.5%	24.9%
肺	66.1%	0.5%	33.4%
食管	60.6%	0.5%	38.9%
胃	55.3%	1.5%	43.2%
结直肠	26.1%	2.5%	71.4%
肝	25.7%	0.5%	73.8%
膀胱	23.5%	0.5%	76.0%
卵巢	22.1%	5.0%	72.9%
肾	20.7%	1.0%	78.3%
胰腺	18.0%	5.0%	77.0%
乳腺	15.1%	1.5%	83.4%
血液	14.3%	0.5%	85.2%
甲状腺	0.5%	1.5%	98.0%
脑	0.2%	0.5%	99.3%
前列腺	0.0%	4.5%	95.5%
整体	28.9%	5.0%	66.1%

66%是整体比率，有些癌症随机因素或许高，但中国高发的几大癌症，则主要是环境和生活方式引起的。（美国杜克大学癌症生物学博士菠萝）

人类大脑越来越大，可能影响大脑的长期健康

- **【背景】** 随着美国人的整体健康状况明显提升，人们的平均寿命也随之增长。越来越多的人会面临阿尔茨海默病及相关痴呆症的挑战。有趣的是，包括著名的弗雷明汉心脏研究（FHS）结果在内的数据显示，**痴呆症发病率实际上有所下降**。
- **【提问】** 研究人员好奇，从生命的早期阶段开始，健康状况、教育水平和血管风险因素管理的**改善是否会促使大脑更好地发育**，进而拥有更大的脑容量。
- **【方法】** 使用的脑部数据来源于1999年至2019年间进行的高分辨率磁共振成像（MRI）扫描，覆盖了从上世纪30年代到70年代出生的人群，反映了广泛的年龄和健康趋势。这次脑容量检查的总样本包括 3226 人，男女比例接近均衡，接受了**MRI以提供大脑结构的详细影像**。
- **【结果】** 研究发现，20世纪70年代出生的人相较于30年代出生者，**脑容量平均增大了6.6%，脑表面积增大近 15%**。即使在对身高和其他因素进行调整后，脑容量的增加仍然显著。
- **【科学意义】** 这些发现**提示了大脑储备增强的可能性**。大脑储备(reserve)指大脑在不表现出认知能力下降的明显症状的情况下，承受衰老或疾病导致的病理损伤的能力。更大的大脑容量和表面积可能会更有效地保护大脑，使其免受老年性疾病的影响。

Kimi帮助解读文献

按下面的条目总结这篇文章：研究背景和科学问题的提出、使用的实验技术和方案、数据质控结果、是否整合了公共组学数据、数据分析和作图亮点、实验验证的方法和结果、主要科学发现或新模型、生物学或临床意义、研究的局限性



JAMA 24 Trends in I...

PDF, 426.16 KB



研究背景和科学问题的提出：

本研究探讨了1930至1970年间出生的Framingham心脏研究参与者的颅内和大脑容量趋势。研究的科学问题在于，是否在这段时间内，由于环境因素的影响，颅内和大脑容量以及皮层厚度出现了时间依赖的长期差异。

使用的实验技术和方案：

研究使用了社区基础的Framingham心脏研究队列的数据，参与者没有痴呆或中风病史，并接受了磁共振成像（MRI）。研究从1999年3月18日至2019年11月15日收集了MRI数据，最终分析数据集创建于2023年10月。

数据质控结果：

研究中使用了多种方法来控制数据质量，包括使用高分辨率3D T1加权MP RAGE成像技术，以及使用稳健的卷积神经网络方法自动移除非大脑元素。此外，使用了neuroHarmonize方法来减少不同MRI机器导致的测量差异。

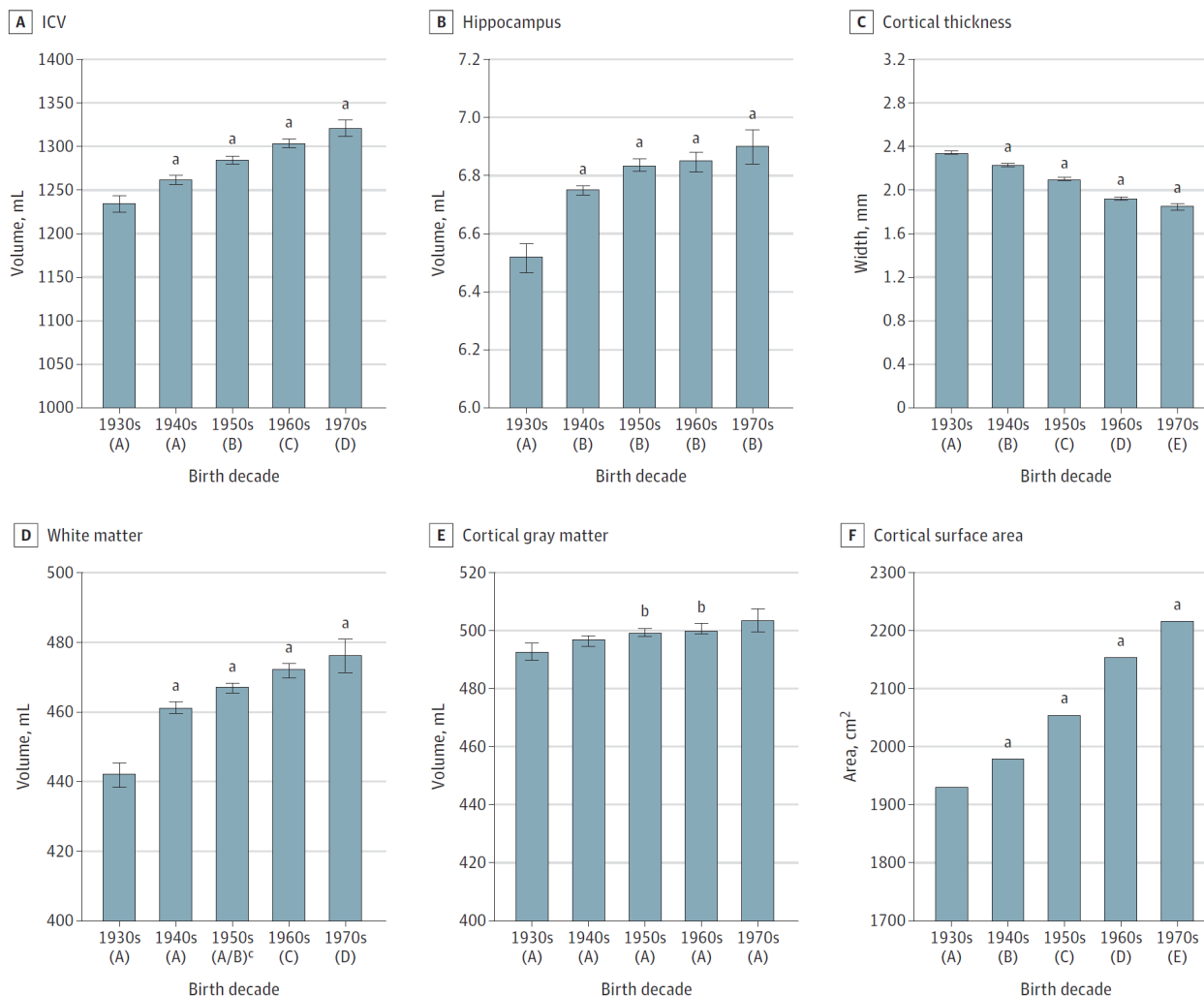


点击链接查看和 Kimi 智能助手的对话

<https://kimi.moonshot.cn/share/crp4gvabhd5g8qftbh2g>

Least-Squares Mean Association (Adjusting for Age and Sex)

Figure 2. Least-Squares Mean Association (Adjusting for Age and Sex) Between Decade of Birth and Magnetic Resonance Imaging Measures



The parenthetical letters A, B, C, D, and E under each decade indicate significant group differences using a Tukey honestly significant differences post hoc comparison. ICV indicates intracranial volume.

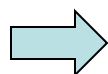
^a $P < .001$ compared with 1930s.

^b $P < .05$ compared with 1930s.

JAMA 24 Trends in
Intracranial and
Cerebral Volumes of
Framingham Heart
Study Participants
Born 1930 to 1970

本节内容

- 线性回归举例



- 线性回归**模型**

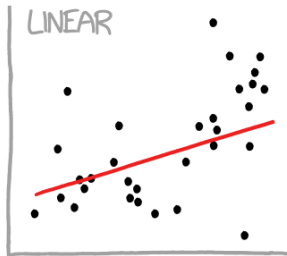
- 统计**陷阱**：均值回归效应

- 统计**作图**：数据分布

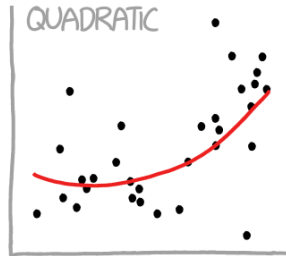
- 助教演示（课后练习）：
 - RStudio、文件读取、作图

如何选择模型？统计知识+建模经验

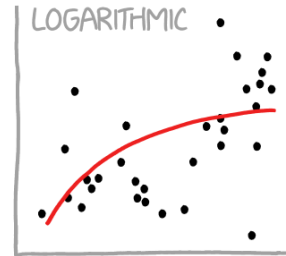
Curve-fitting methods & messages



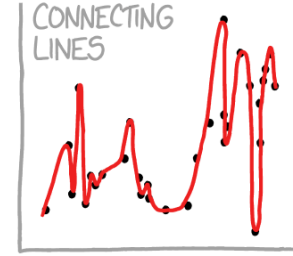
"HEY, I DID A REGRESSION."



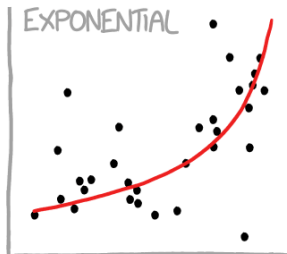
"I WANTED A CURVED LINE, SO I MADE ONE WITH MATH."



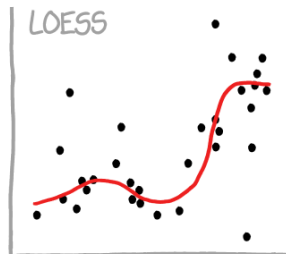
"LOOK, IT'S TAPERING OFF!"



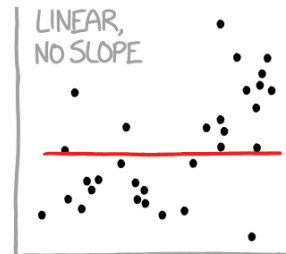
"I CLICKED 'SMOOTH LINES' IN EXCEL."



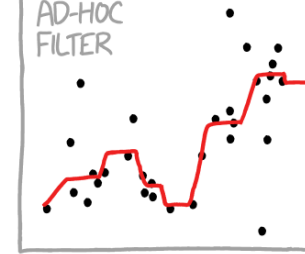
"LOOK, IT'S GROWING UNCONTROLLABLY!"



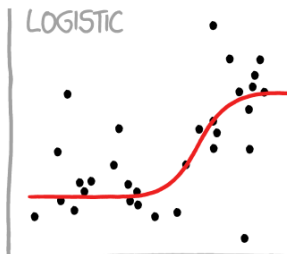
"I'M SOPHISTICATED, NOT LIKE THOSE BUMBLING POLYNOMIAL PEOPLE."



"I'M MAKING A SCATTER PLOT BUT I DON'T WANT TO."



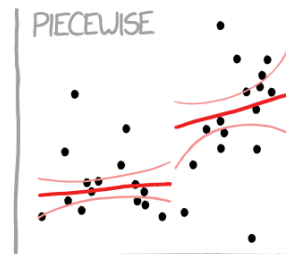
"I HAD AN IDEA FOR HOW TO CLEAN UP THE DATA. WHAT DO YOU THINK?"



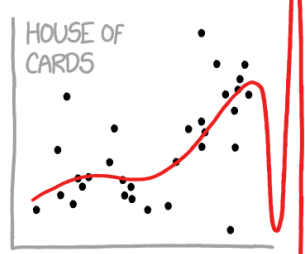
"I NEED TO CONNECT THESE TWO LINES, BUT MY FIRST IDEA DIDN'T HAVE ENOUGH MATH."



"LISTEN, SCIENCE IS HARD. BUT I'M A SERIOUS PERSON DOING MY BEST."



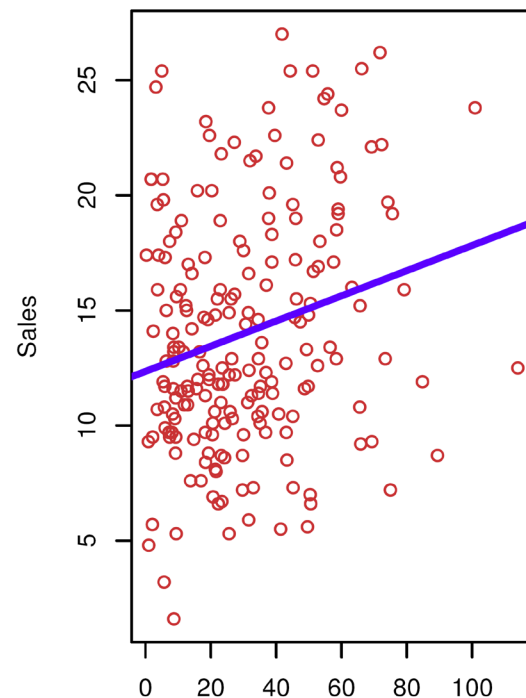
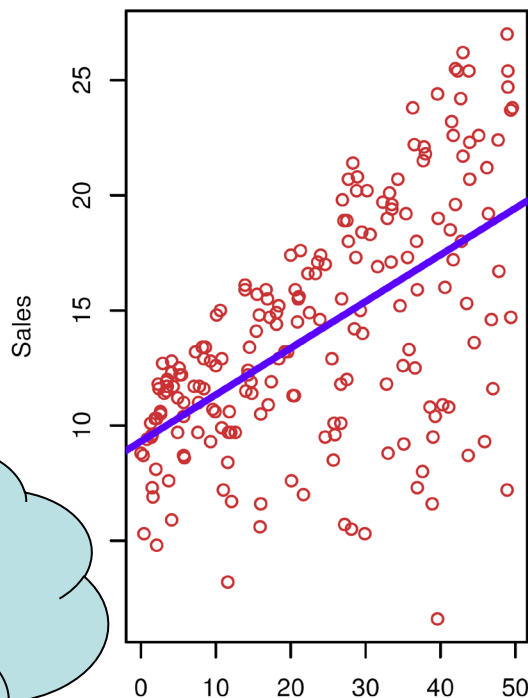
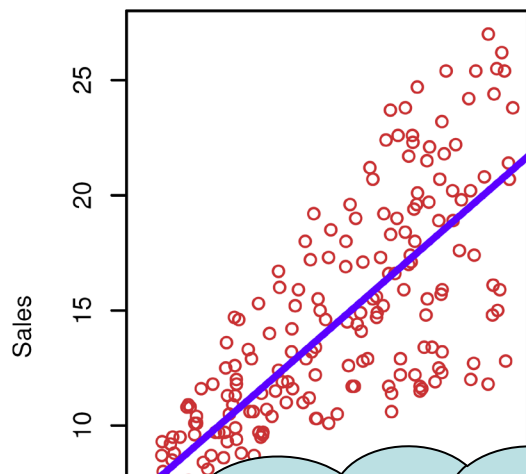
"I HAVE A THEORY, AND THIS IS THE ONLY DATA I COULD FIND."



"AS YOU CAN SEE, THIS MODEL SMOOTHLY FITS THE- WAIT NO NO DON'T EXTEND IT AAAAAA!!"

Advertising数据集

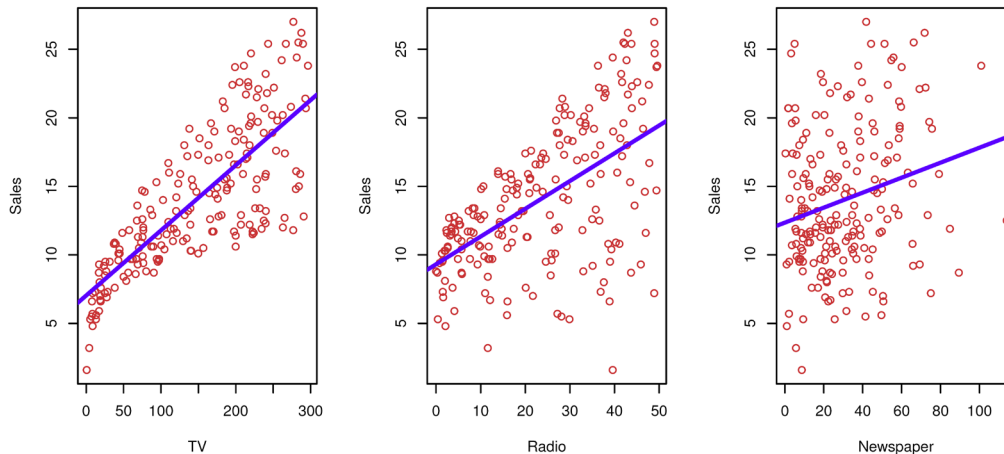
	TV	Radio	Newspaper	Sales
1	230.1	37.8	69.2	22.1
2	44.5	39.3	45.1	10.4
3	17.2	45.9	69.3	9.3
4	151.5	41.3	58.5	18.5



开发基于三类广告
媒体预算的精准预
测销量的模型？

Advertising营销问题

- 广告预算和销量有关吗？
- 这种关系有多强？
- 哪种媒体能促进销售？
- 如何精确地估计每种媒体对销量的影响？
- 对未来销量的预测精度如何？
- 预测关系是否是线性的？
- 广告媒体之间是否存在协同效应？



统计模型

$$Y = f(X) + \epsilon$$

$$X = (X_1, X_2, \dots, X_p)$$

- f : 固定但未知, 表达了 $\mathbf{X}$ (预测变量) 提供给 $\mathbf{Y}$ (响应变量) 的**系统、确定** (systematic) 信息
- ϵ : **随机误差项** (error term) , 假设与 X 独立、均值为0
- **统计 (机器) 学习**是估计 f 的一系列方法, 并对**估计的优劣**进行评价

估计 f 的用处1：预测

- 预测 (prediction)

$$\hat{Y} = \hat{f}(X)$$

- $\hat{f}$ 是黑箱，准确地预测 Y ，不追求它的确切形式
- 例子： X 是病人的血样特征， Y 测量病人疾病风险（液体活检）、使用药物后不良反应的风险
- **可约误差**：当 $\hat{f}$ 不是 f 的最佳估计时、优化 $\hat{f}$
- **不可约误差**： Y 还是一个关于 ε 的函数， ε 不能用 X 预测，可能包括没有观测的变量信息

不良反应的风险在一天内不同；药效与服药温度有关

估计 f 的用处2：推断

- 推断 (inference)

- 对 $X_1, \dots, X_p$ 变化时如何影响 Y 感兴趣。不全是为预测 Y , 而要理解 X 和 Y 的关系

- 哪些预测变量与响应变量相关?

- 响应变量和每个预测变量之间的关系?

- 这种关系是线性的吗, 还是需要更复杂的形式描述?

- 哪类媒体对销量有直接贡献、产生最大促进作用?
 - 如果增加一定电视广告费用, 对销量能带来多少增加?

- **预测: X 不可变** (如预测日食), **推断: X 可控** (做课后练习->数据分析能力强)

简单线性回归：模型

模型：

$$Y \approx \beta_0 + \beta_1 X$$

$$\text{sales} \approx \beta_0 + \beta_1 \times \text{TV}$$

画图是线性关系

预测：

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

最小化残差平方和：

$$e_i = y_i - \hat{y}_i$$

$$\text{RSS} = e_1^2 + e_2^2 + \cdots + e_n^2$$

Residual Sum Squares

最小二乘估计：

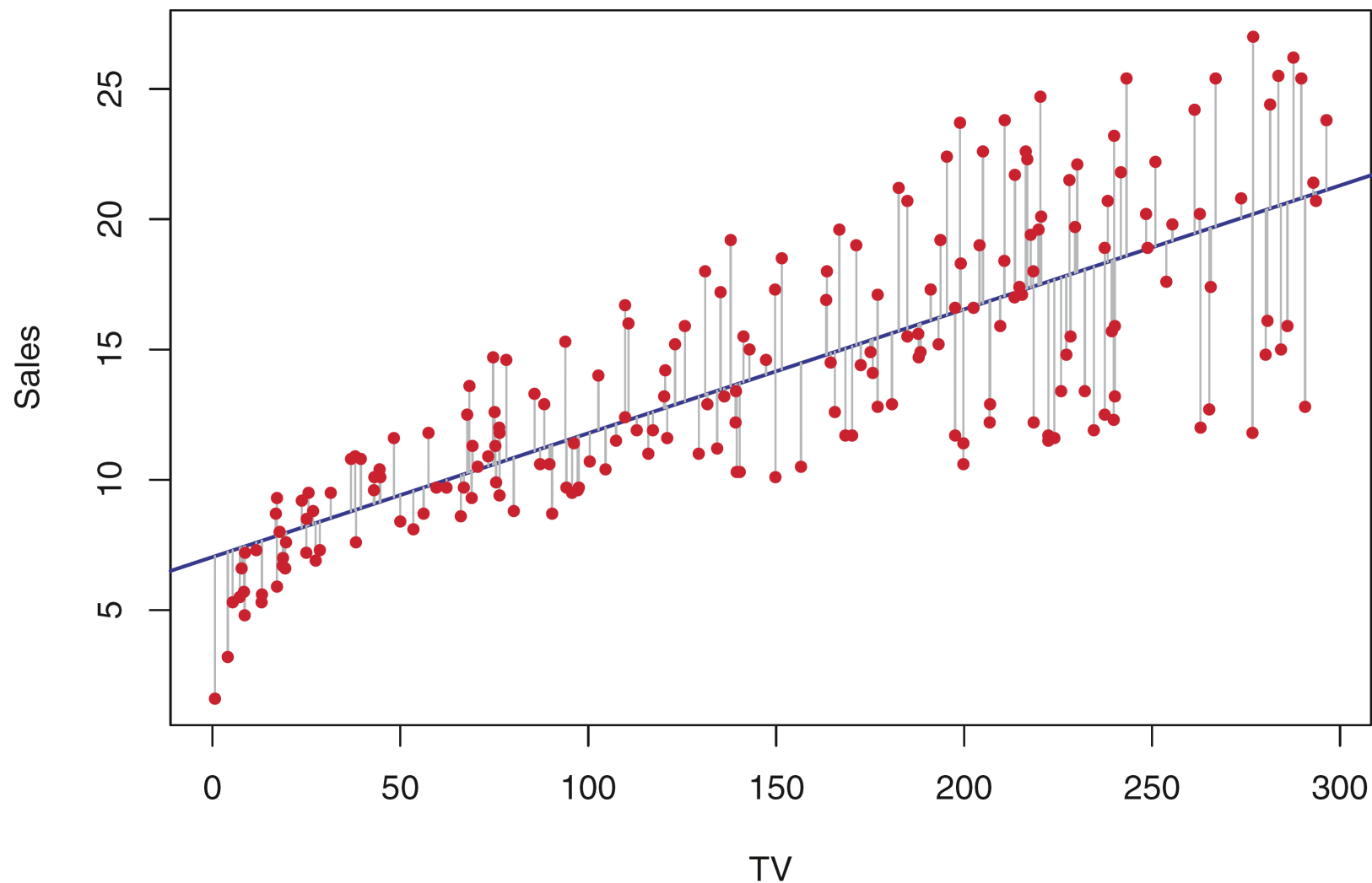
$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2},$$

如何使它为正值？

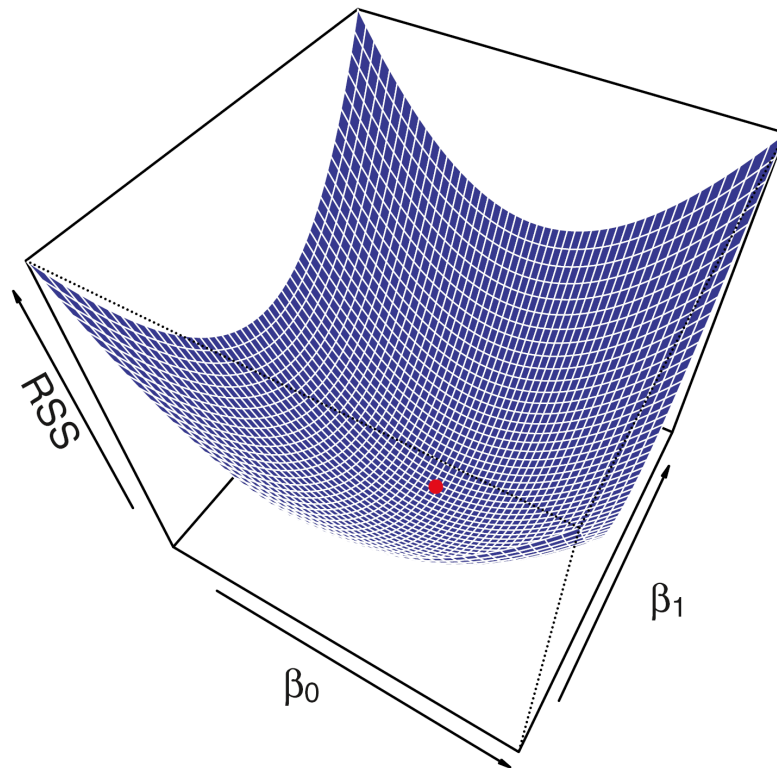
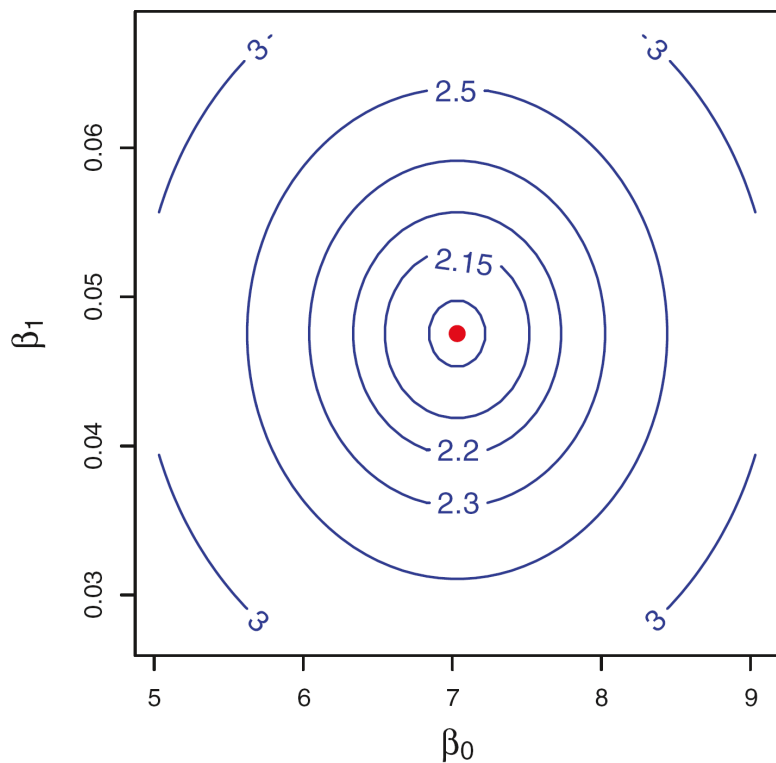
$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x},$$

回归线穿过 (x均值, y均值) 点

简单线性回归：拟合图



简单线性回归：RSS轮廓图



能找到全局
唯一最优解

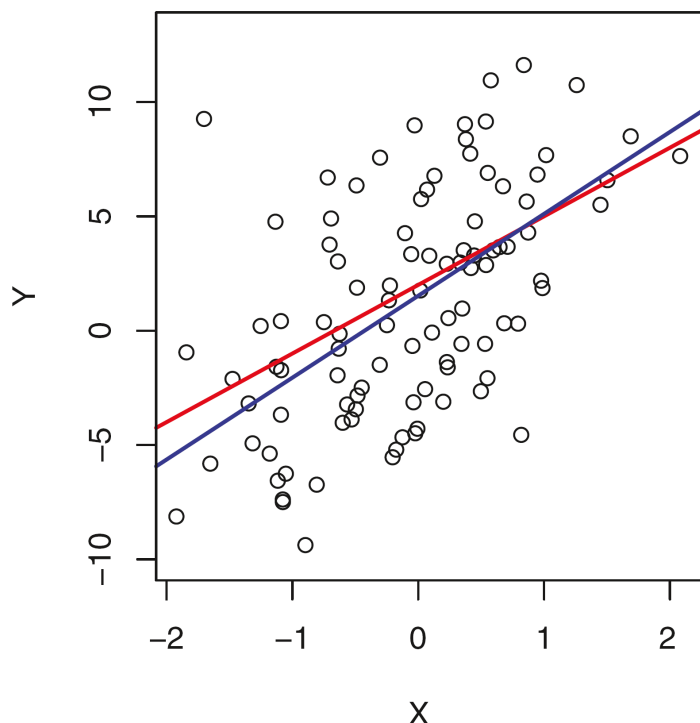
评价参数估计的准确性：模拟数据

红线： 真实关系，用以模拟产生数据点

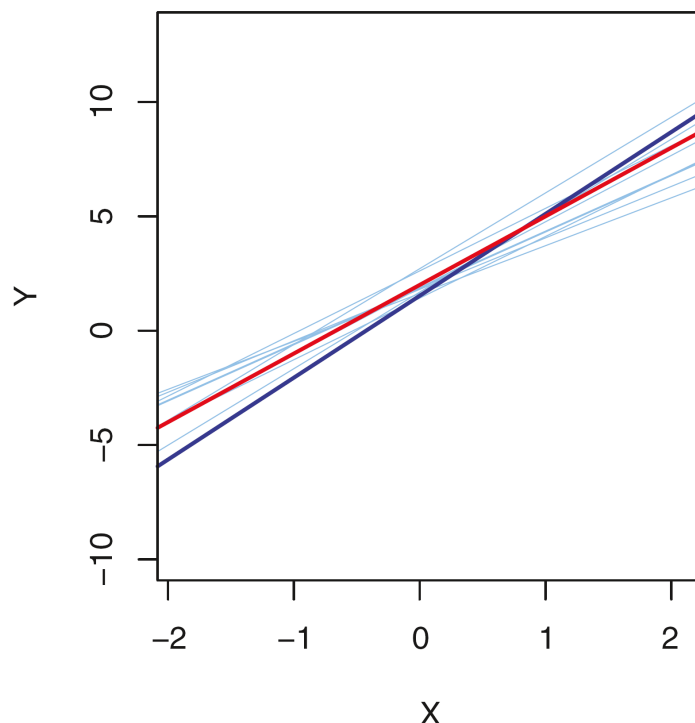
$$Y = 2 + 3X + \epsilon$$

蓝线： 基于模拟数据点的回归线

统计思维： 模拟产生金标准数据



模拟一次数据



模拟多次数据

- 不同的模拟数据估计出不同的 β 值（浅蓝线）
- β 估计的**无偏性**：不会系统地高估或低估真实参数 β ： $E(\hat{\beta}) = \beta$

评价参数估计的准确性：标准误差

(Standard Error, 参数标准差的估计值)

$$SE(\hat{\beta}_0)^2 = \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right], \quad SE(\hat{\beta}_1)^2 = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\sigma^2 = \text{Var}(\epsilon)$$

$$Y = \beta_0 + \beta_1 X + \epsilon$$

σ 估计值: $RSE = \sqrt{RSS/(n-2)}$
残差标准误, residual standard error

理论vs.直观: 当 x 的取值更分散时, SE取值更小

模型假设: 需要每个 ϵ 独立, 有相同的方差 (在残差图中是否成立?)

第一节课例子: Buffon's needle例子: 针 (样本) 越多, 估计的 π 越准确

对参数进行假设检验

- 零假设 (H_0) : X 和 Y 之间没有关系

$$\beta_1 = 0, Y = \beta_0 + \varepsilon$$

做t-检验：需要
error正态分布的
模型假设

- 备选假设 (H_a) : X 和 Y 之间有一定的关系

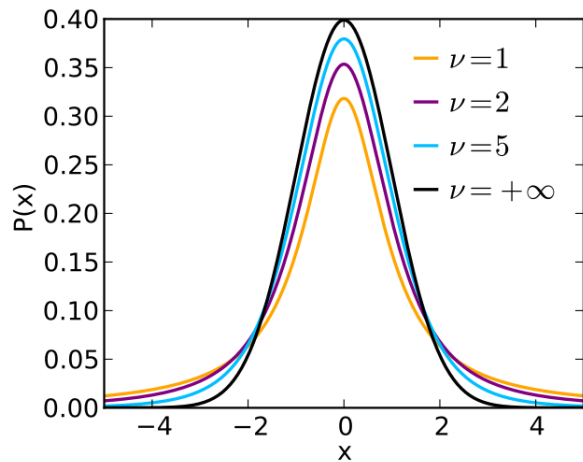
$$\beta_1 \neq 0, Y = \beta_0 + \beta_1 X + \varepsilon$$

- β_1 是否是0? 在零假设下, t 统计量比较 β_1 估计值的绝对值vs.准确性 (标准误差), 服从 t 分布:

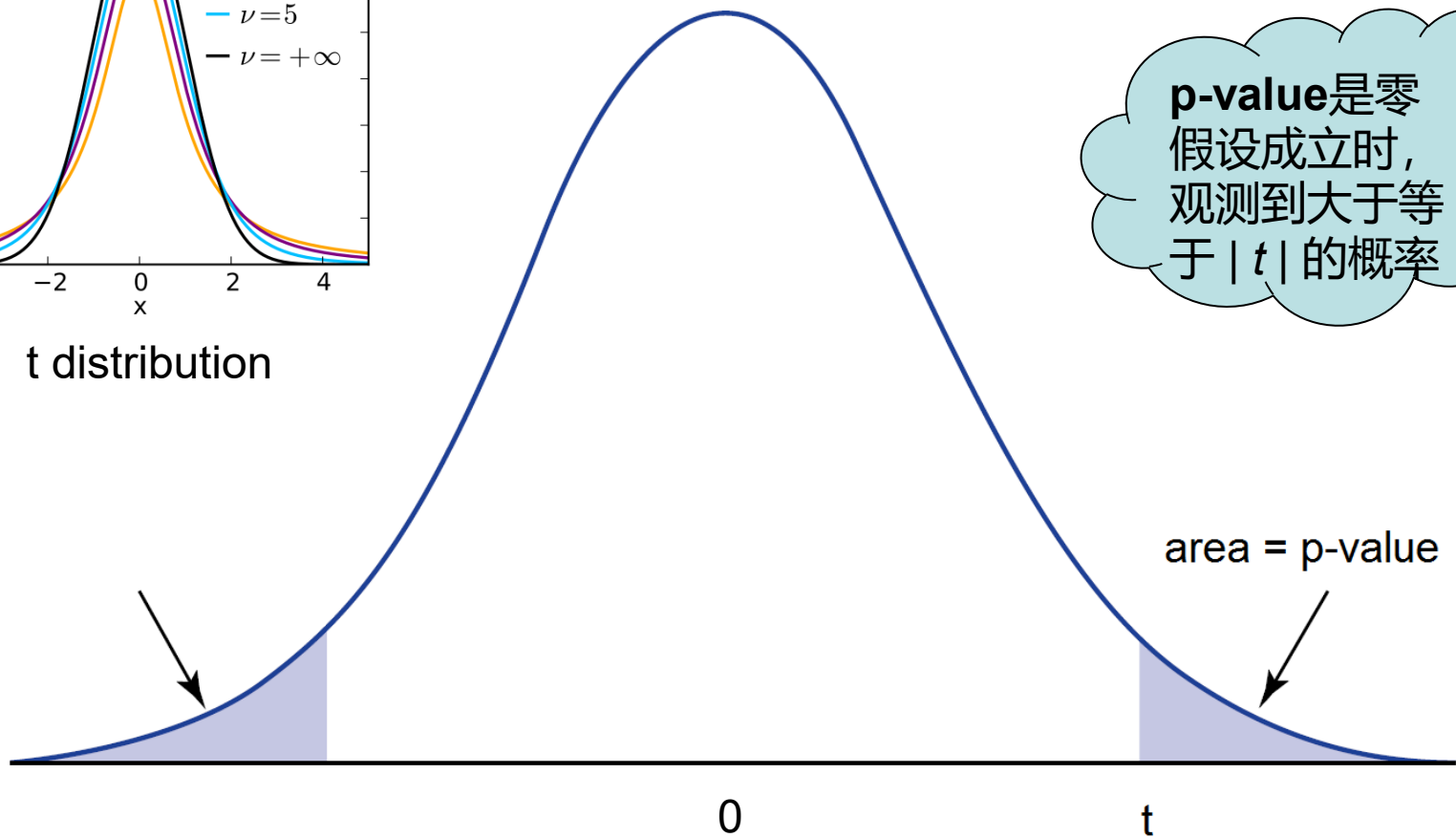
$$t = \frac{\hat{\beta}_1 - 0}{\text{SE}(\hat{\beta}_1)}$$

		Coefficient	Std. error	t-statistic	p-value
β_0	Intercept	7.0325	0.4578	15.36	< 0.0001
β_1	TV	0.0475	0.0027	17.67	< 0.0001

From t-statistic to p-value



t distribution



p-value是零
假设成立时,
观测到大于等
于 $|t|$ 的概率

The distribution of the t-statistic under the null hypothesis

量化模型拟合数据的程度

解读思路： R^2 的解释经常出现在文献摘要或标题中

- R^2 统计量

$$R^2 = \frac{\text{TSS} - \text{RSS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}} \quad \text{TSS} = \sum (y_i - \bar{y})^2$$

- TSS (总平方和)：响应变量 Y 的总体差异性 (variation)
- RSS：进行回归后仍然无法解释的 Y 的差异性
- TSS – RSS：响应变量进行回归后被解释 (消除) 的差异性
- R^2 ：Y 的差异性能被 X 解释的部分所占的比例 (0.61)

统计思维

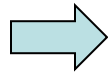
- 数据是按照什么模型产生的？噪音、偏差如何产生的？
- 如何评价一个模型拟合数据的好坏？
- 预测未来时如何判断准确度、避免过拟合？

课后练习：结合简单线性模型的案例回答上述问题

本节内容

- 线性回归举例

- 线性回归**模型**



- 统计**陷阱**：均值回归效应

- 统计**作图**：数据分布

- 助教演示（课后练习）：

- RStudio、文件读取、作图

卡尔·皮尔逊 (Karl Pearson)



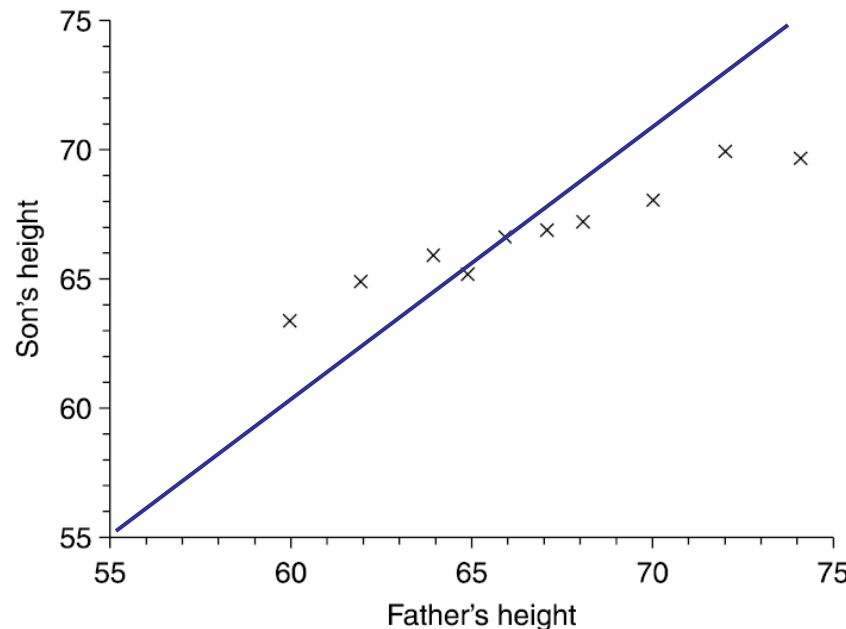
1857 ~ 1936

- 提出卡方 (χ^2) 检验
- 发展了相关性和回归理论
- 重视个体变异性的数量表现和变异数据的处理
(1894年提出了“标准差”及其符号 σ)
- 美国生物学家雷蒙德·珀尔说：“历史的进展清楚地显示出有极少数的人，没有靠地位或者环境，而仅仅靠他们自己的聪明才智，对人类的思维模式产生了巨大的影响……卡尔·皮尔逊就是这少数杰出人物中的一个。”

Stat Pitfalls:

Regression to the mean (均值回归效应)

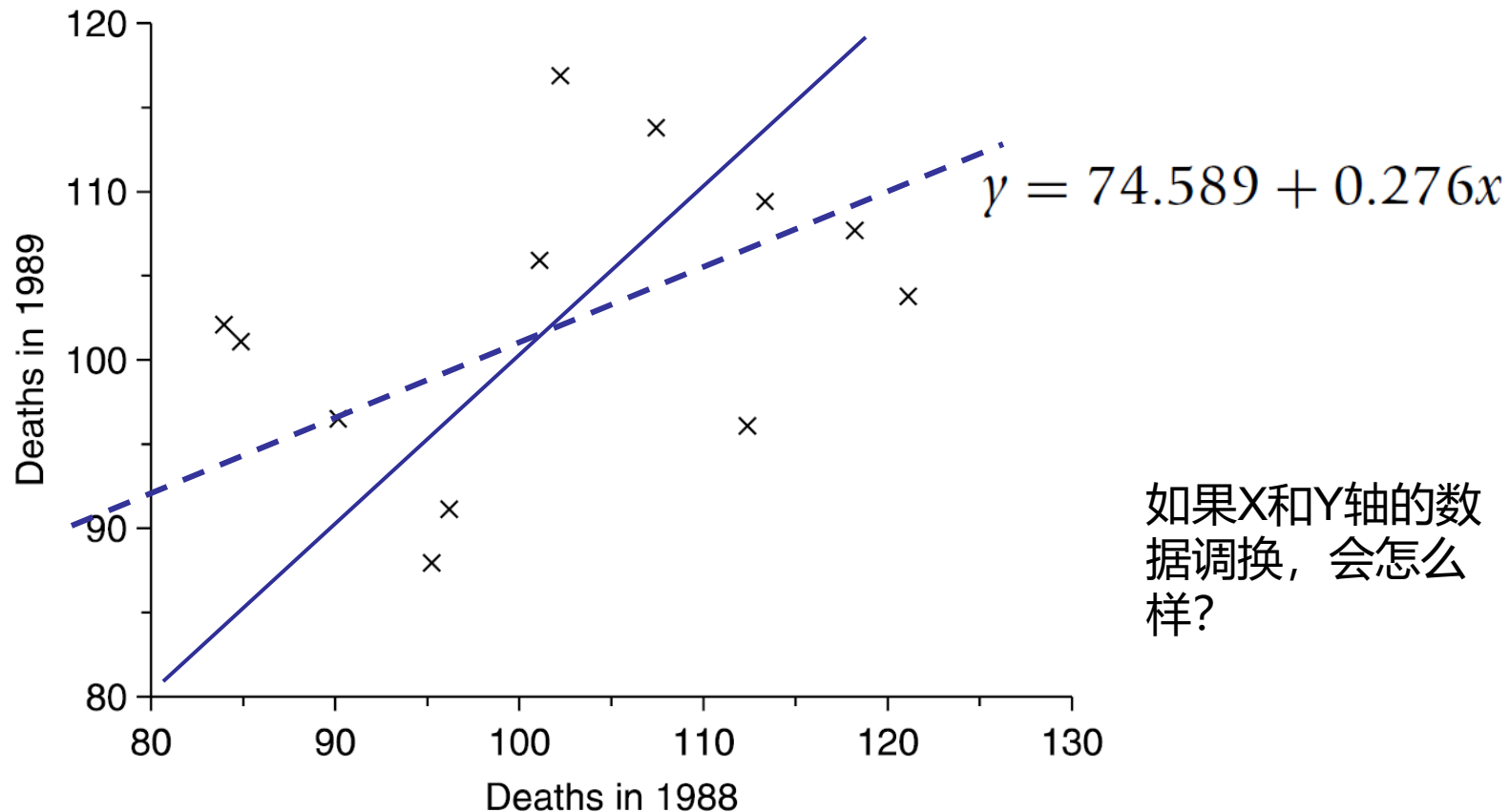
Father's height	Son's height	Father's height	Son's height
60	63.6	67	67.1
62	65.2	68	67.4
64	66	70	68.3
65	65.5	72	70.1
66	66.9	74	70



Karl Pearson's data

Sons of fathers who are either extremely short or extremely tall tend to be more “average” than their fathers

Regression to the mean: motor vehicle accidents in different counties



如果X和Y轴的数据调换，会怎么样？

FIGURE 12.6

Scatter diagram of 1989 deaths versus 1988 deaths.

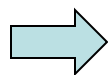
当心“回归谬误”

- 为什么电影续集常常让人失望？
 - 《唐伯虎点秋香2：四大才子》
- 为什么每况愈下的病情似乎被某种治疗方法“治好”了？
- 为什么一败再败的球队换教练之后成绩反弹？
 - “盛极必衰”，“物极必反”，正是统计回归效应的表现。
 - 不理解回归平均，就很容易犯回归谬误，错误地从回归平均效应中得出因果关系
 - 健康领域的众多伪科学、巫术、骗局令患者深信不疑，靠的就是“回归谬误”
- 如何避免回归谬误？在健康领域，治疗的科学标准要求“循证”，即疗效要有证据支持（随机分组、对照实验）

http://news.ifeng.com/a/20141214/42713024_0.shtml

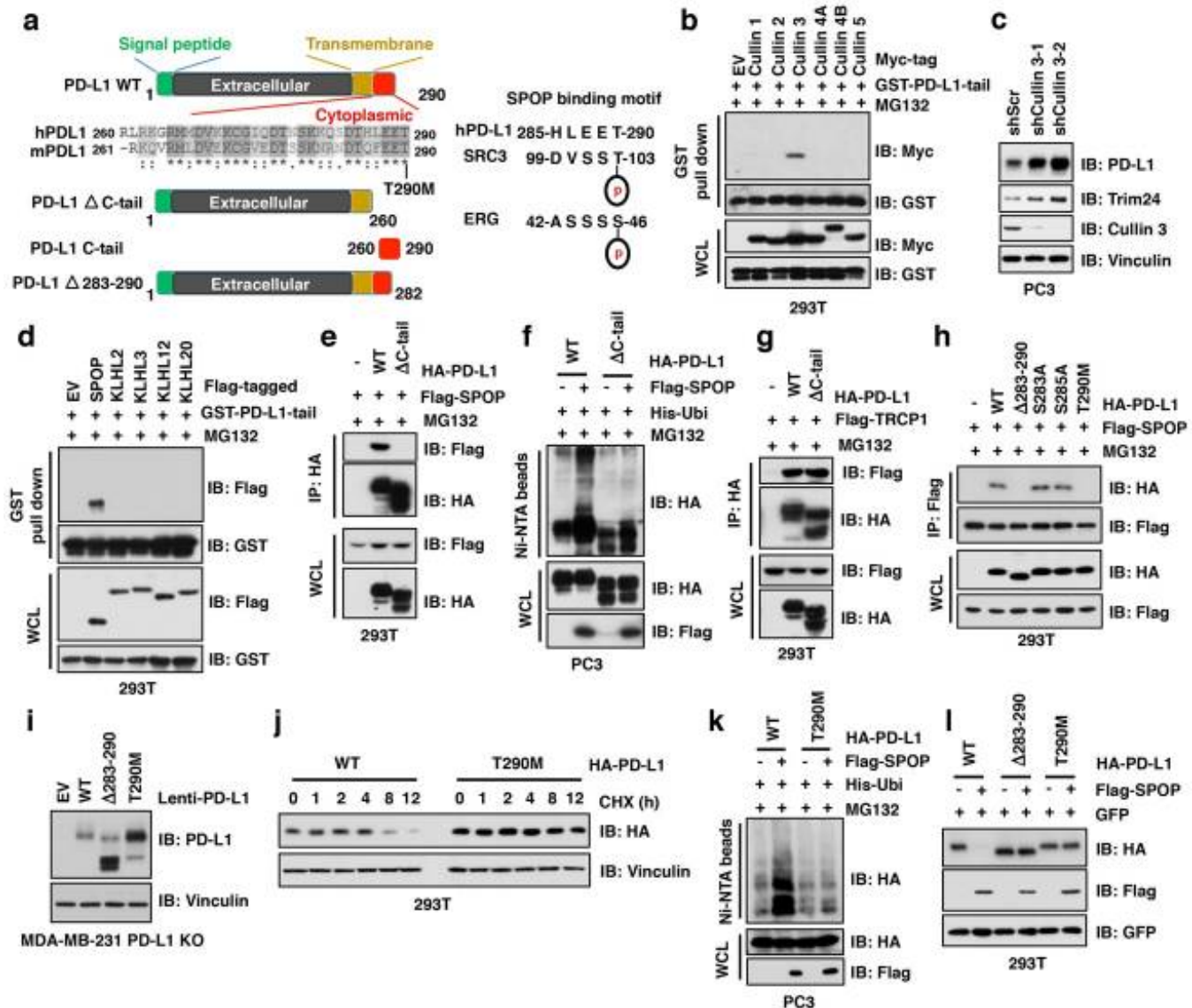
本节内容

- 线性回归举例
- 线性回归**模型**
- 统计**陷阱**：均值回归效应



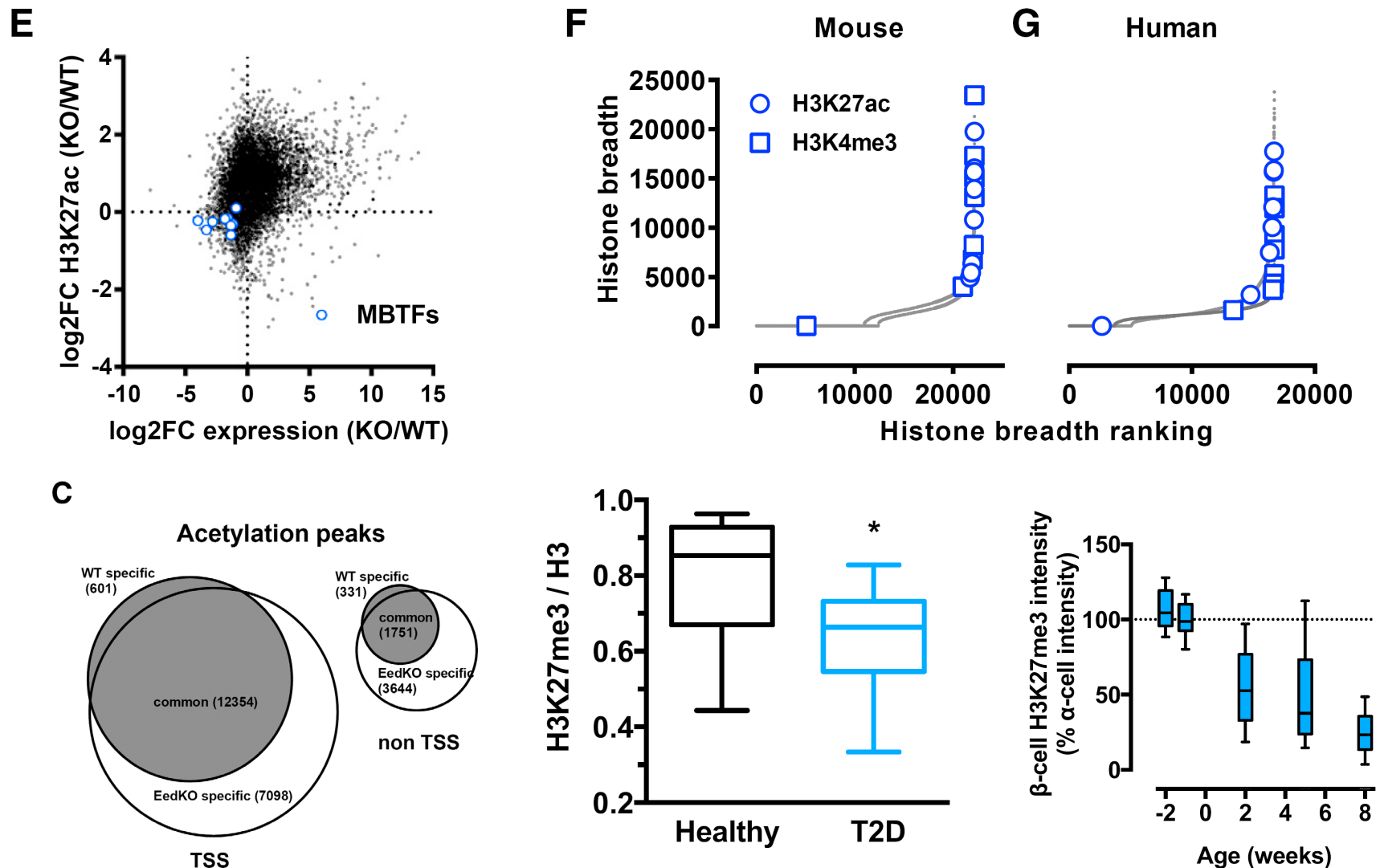
- 统计**作图**：数据分布
- 助教演示（课后练习）：
 - RStudio、文件读取、作图

Traditional figures in a biology paper



Nature 18 Cyclin D-CDK4 kinase destabilizes PD-L1 via cullin 3-SPOP to control cancer immune surveillance

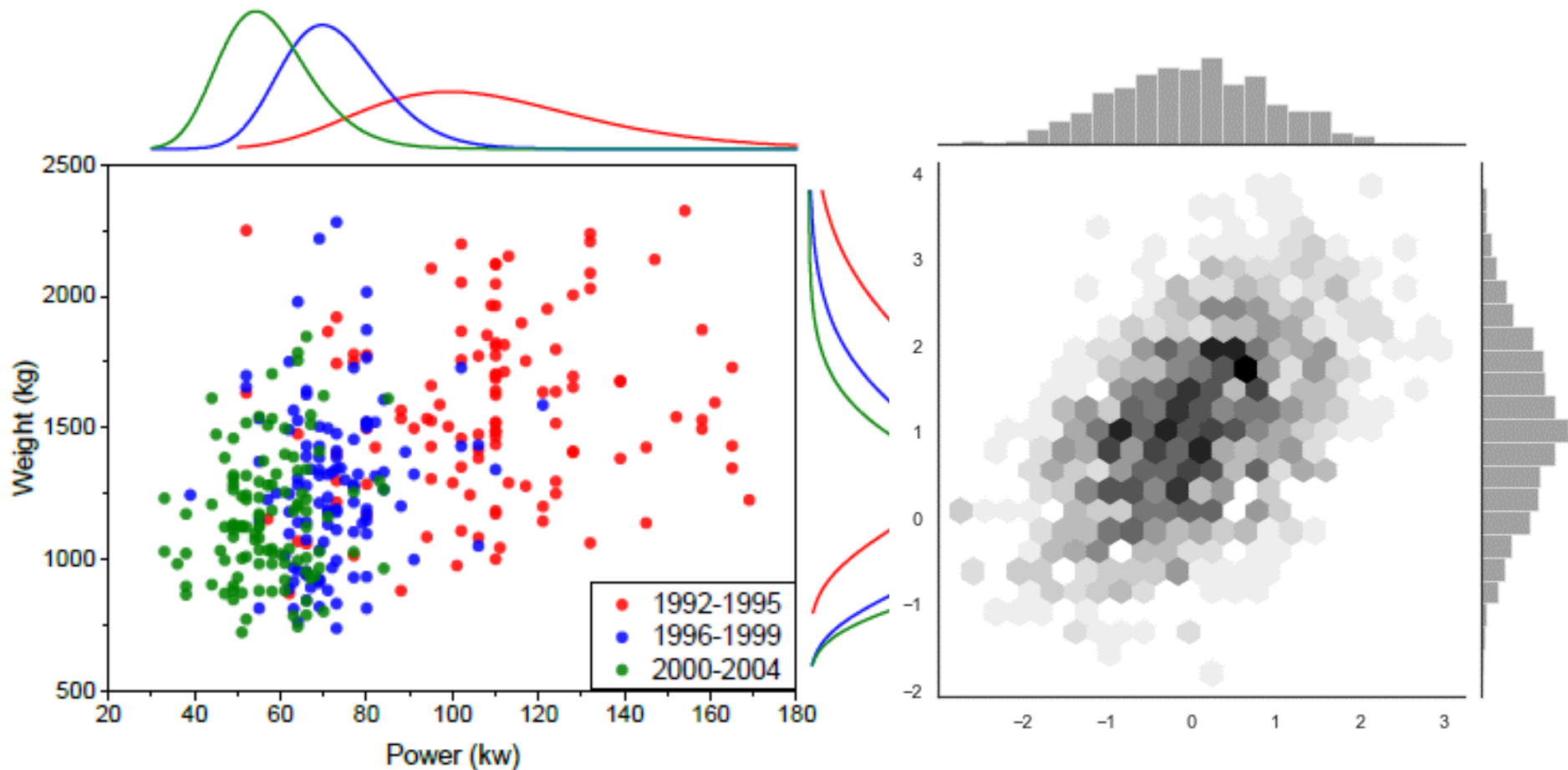
Genomic figures in a biology paper



Genomic figures in a biology paper

- 上页这篇文章还有很多其他类型的图，见PDF文件
- 基因组学数据分析和做图是当代生物学不可或缺的技能
- 通过学习和练习，能做出大部分课题中需要的组学图
- 熟悉这些做图传递的信息（读图能力），也能更快、更准确地理解作者表达的意思，判断数据是否支持作者的结论

Stat graphs: data distribution

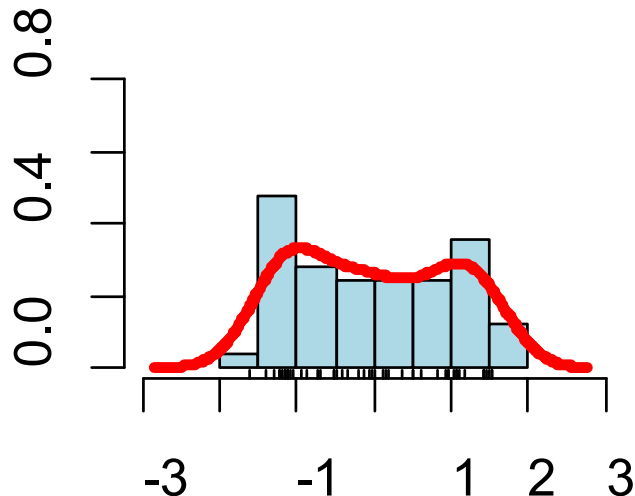


<https://seaborn.pydata.org/tutorial/distributions.html>

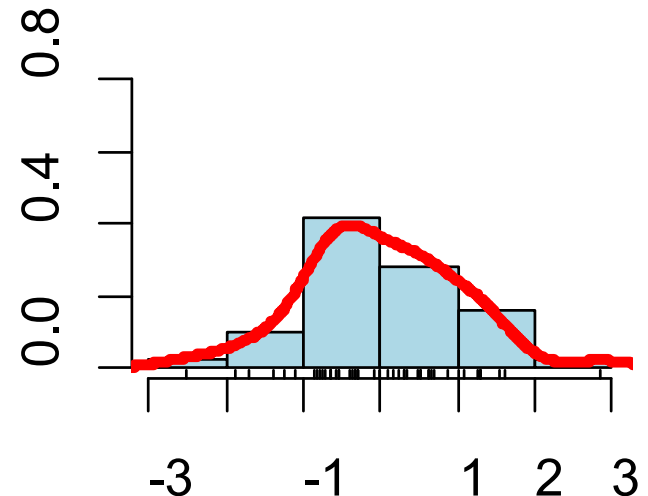
R function: `hist()`, `density()`

Histogram (直方图) to check or compare distributions

Uniform distrib

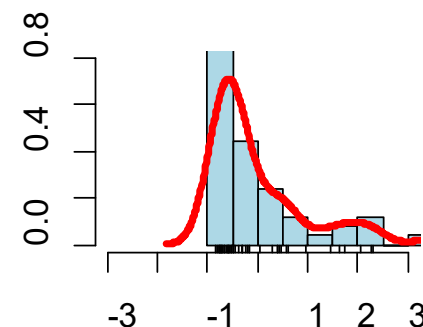


Fat-tailed distr



Y轴：有时直接列出落在每个区间的数的个数，这里进行了归一化，使面积为1，便于和密度图（对直方图拟合曲线）重合

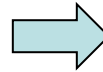
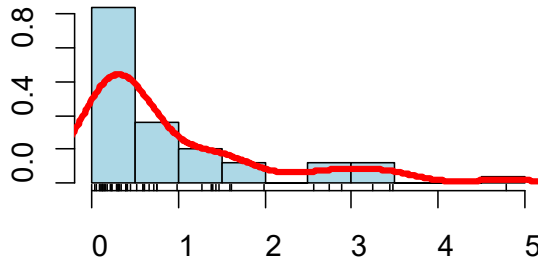
Skewed distr



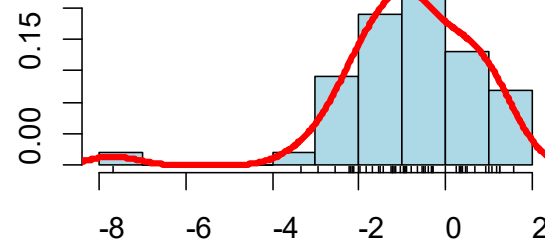
http://zoonek2.free.fr/UNIX/48_R/03.html

Log transformation

Skewed distribution

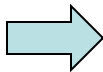
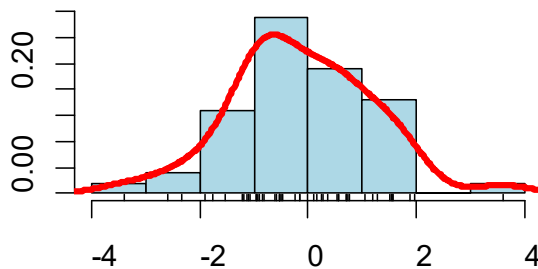


Skewed distribution

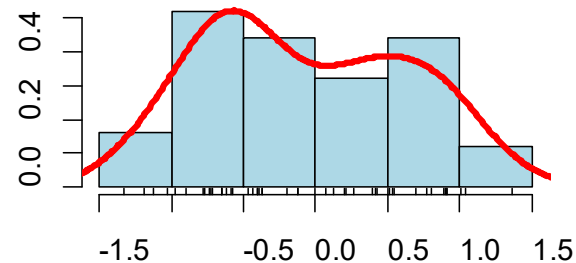


Log normal?

Fat tailed distributio



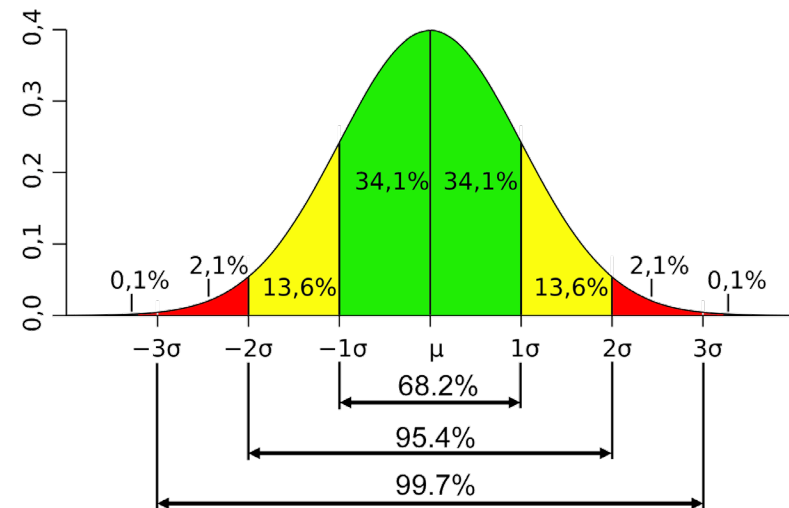
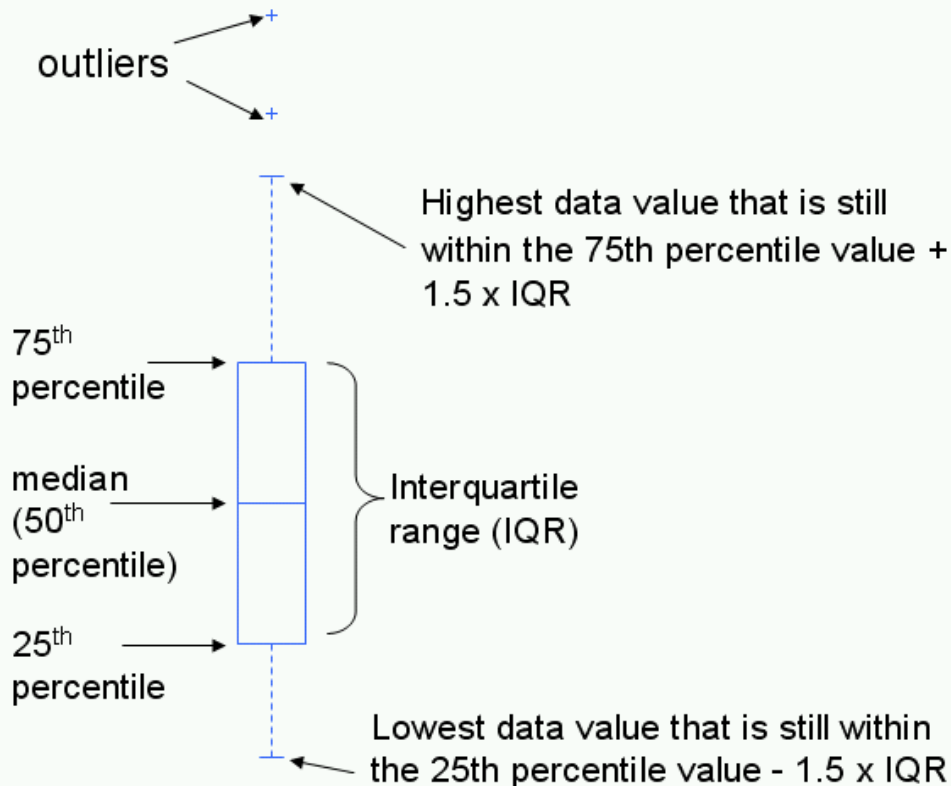
Fat tailed distributio



http://zoonek2.free.fr/UNIX/48_R/03.html

Boxplots (箱线图)

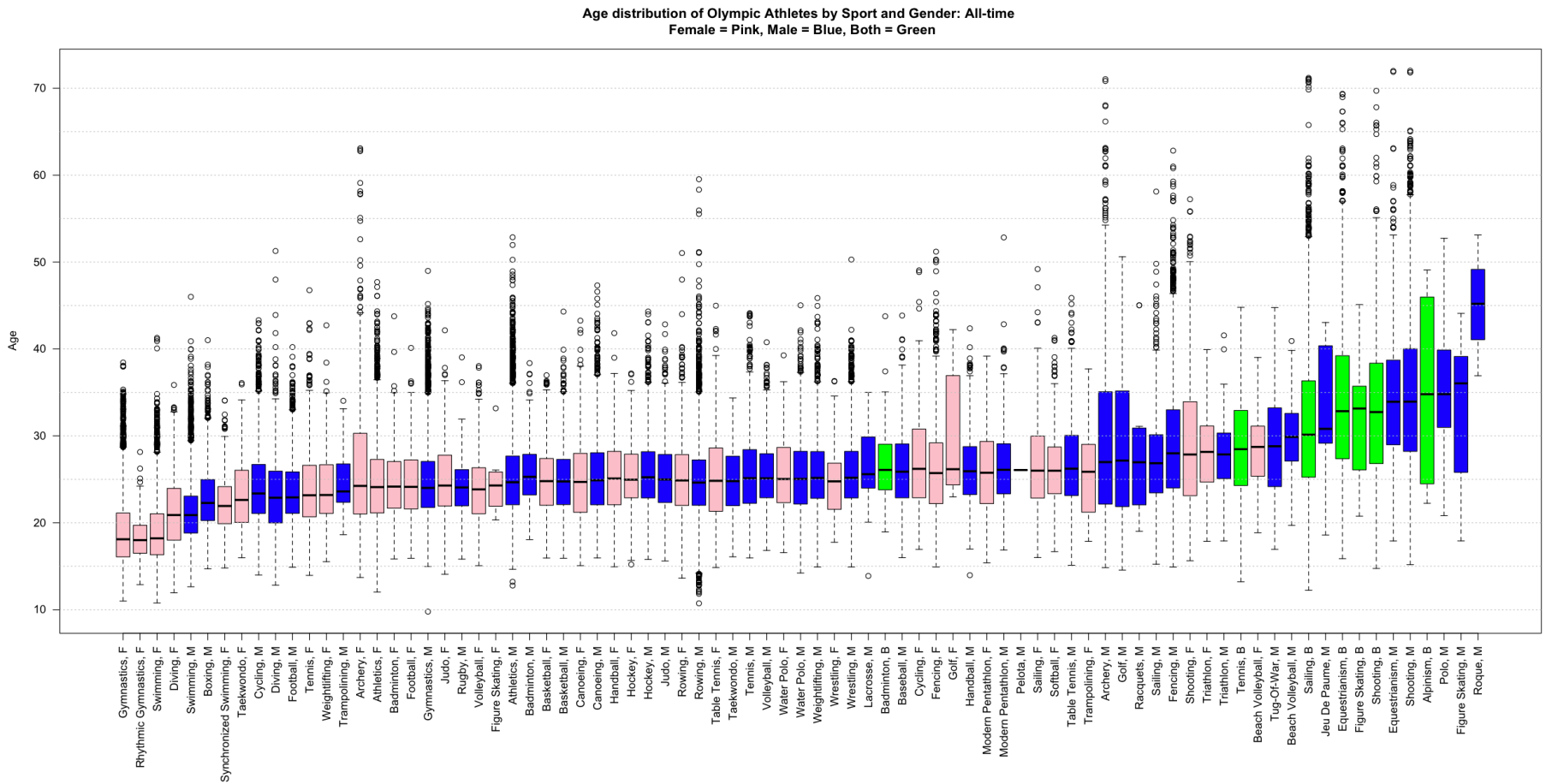
- Box-and-whiskers plots, a graphical summary of a distribution



课后练习: 如果数据来自一个标准正态分布, 画成boxplot后, 离群点的比例大概是多少? 可使用 R 函数 `qnorm()`, `pnorm()` 计算概率, 或 `rnorm()` 进行模拟。

Sorted boxplots: age of Olympics athletes

side-by-size boxplots of age for each sport by gender with blue for male, pink for female, and green for mixed competition.



<http://statsinthewild.com/2012/07/09/olympics-boxplot/>

Violin plot

R packages: [vioplot](#), [wvioplot](#), [caroline](#), [UsingR](#), [lattice](#), [matplotlib](#), and [ggplot2](#) (http://en.wikipedia.org/wiki/Violin_plot)

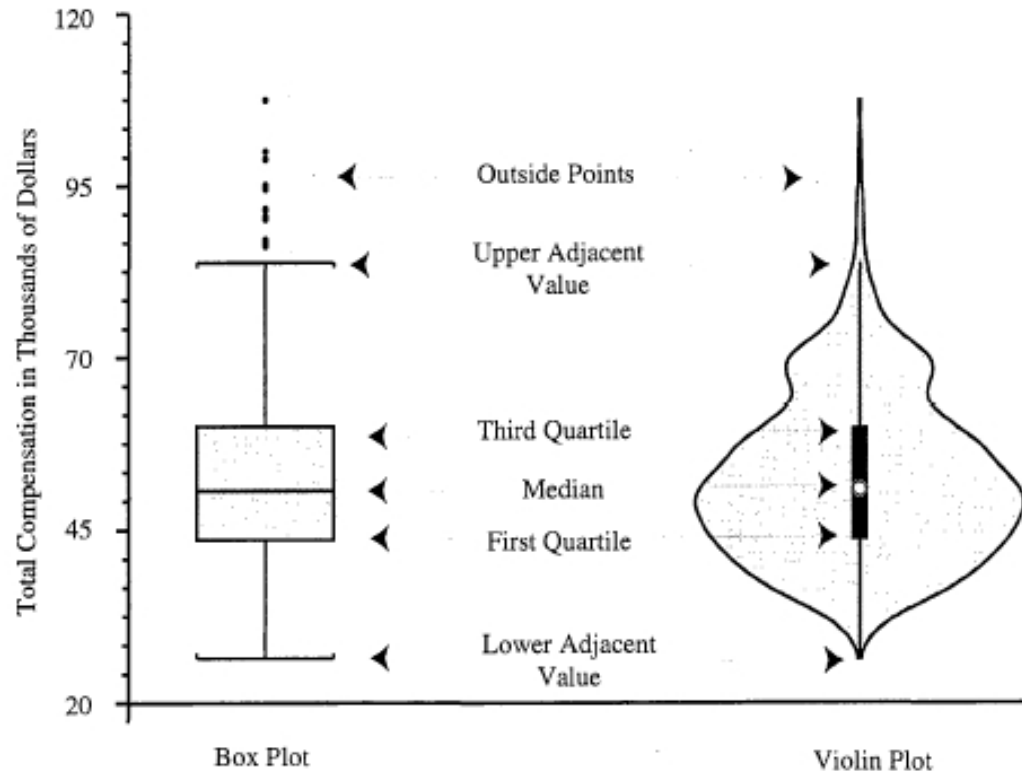


Figure 1. Common Components of Box Plot and Violin Plot. Total compensation for all academic ranks.

<http://www.sci.utah.edu/~kpotter/Library/Papers/hintze:1998:VPDT/>

Comparison of different plots

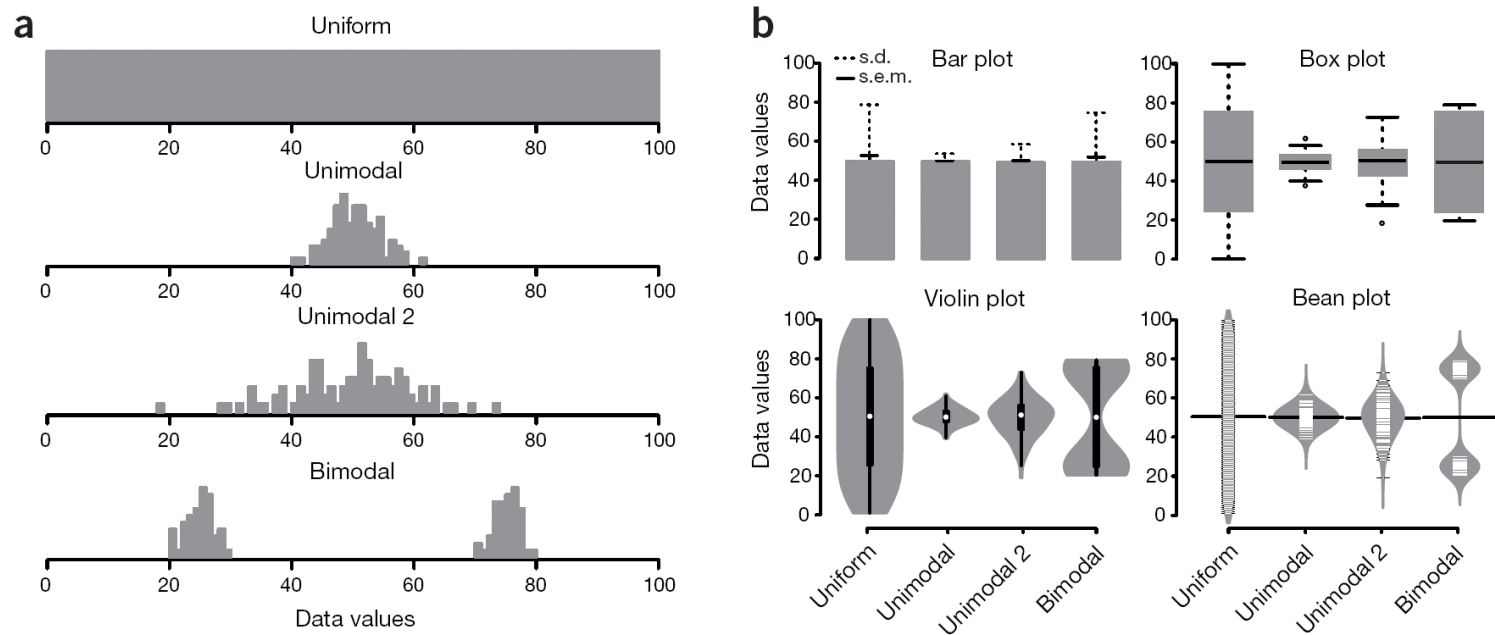


Figure 1 | Data visualization with box plots. (a) Hypothetical sample data sets of 100 data points each that are uniform, unimodal with one of two different variances or bimodal. Simple bar plot representations and statistical parameters may obscure such different data distributions. (b) Comparison of data visualization methods. Bar plots typically represent only the mean and s.d. or s.e.m. Box plots visualize the five-number summary of a data set (minimum, lower quartile, median, upper quartile and maximum). Violin and bean plots represent the actual distribution of the individual data sets.

展示了Bar plot、Box plot、Violin plot和Bean plot对数据分布的反应。从Bar plot上只能看到数据标准差或标准误不同；Box plot可以看到数据分布的集中性不同；Violin plot和Bean plot展示的是数据真正的分布，尤其是对Bimodal数据的展示

<https://www.jianshu.com/p/5682e923128b>

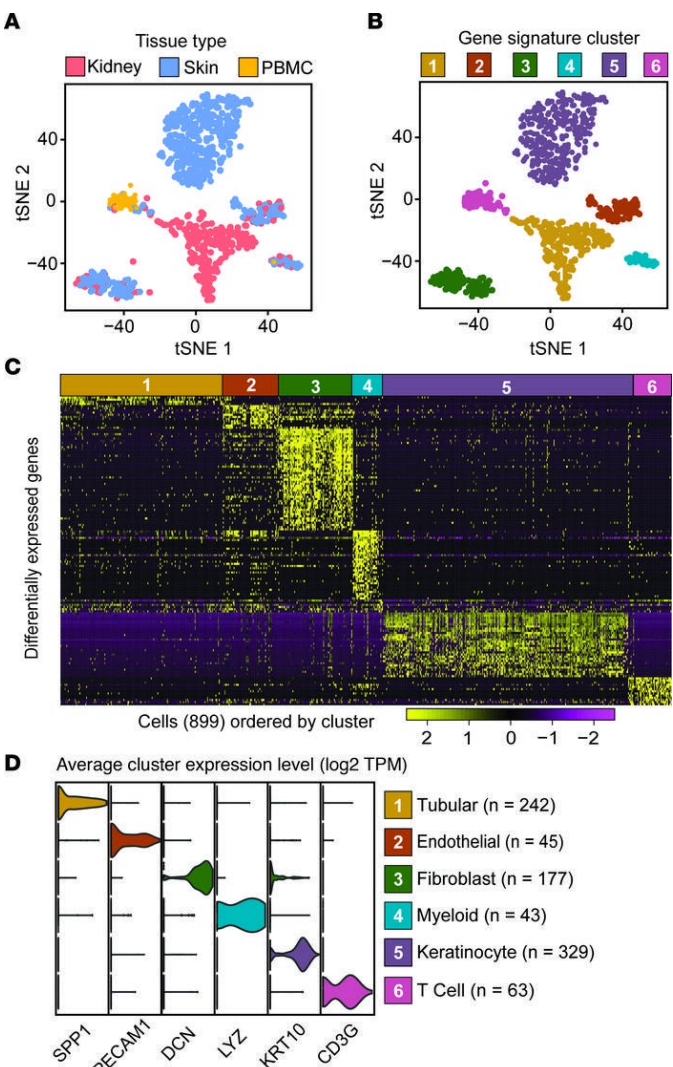
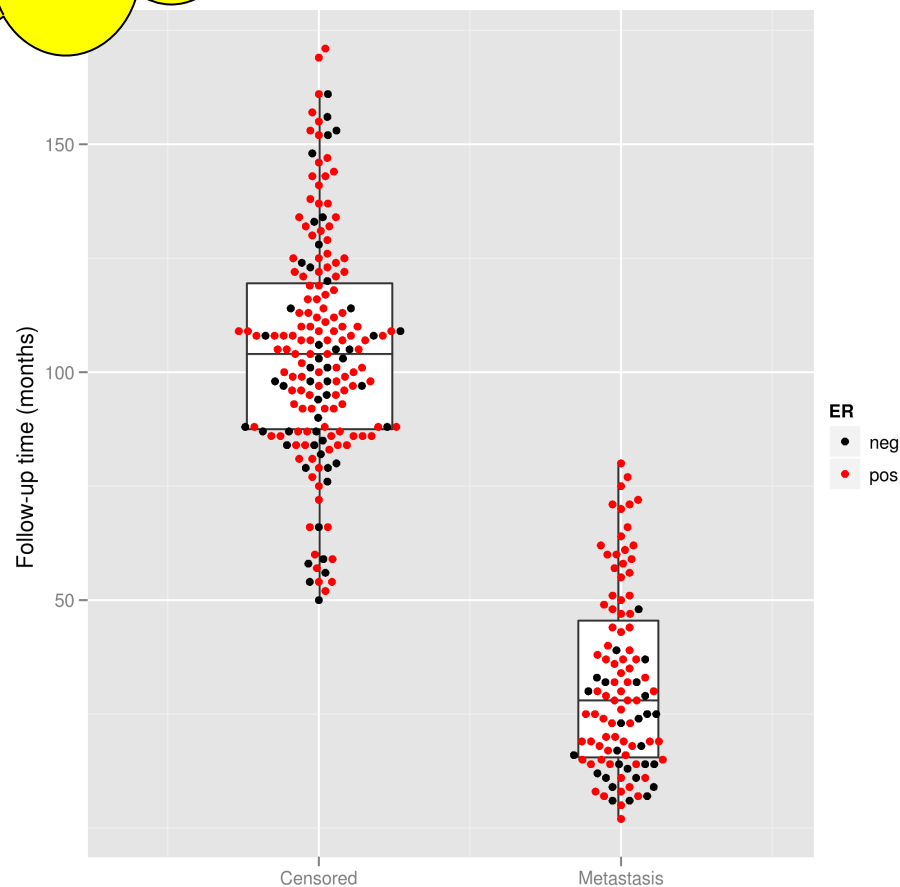
NM 14 BoxPlotR a web tool for generation of box plots

作图思路：
读者喜欢看到有创意的数据图

Be inventive

Combine and connect

Beeswarm plot



<https://denishaine.wordpress.com/2011/09/16/beeswarm-plot-with-ggplot2/beeswarm-ggplot/>

<https://insight.jci.org/articles/view/93009/figure/3>

R graphs: start from simple cases

```
x <- seq(0, 1, .05)
plot(x, x, ylab="y", type="l")
for (i in 2:8)
  lines(x, x^i, col=i, lwd=i)
```

```
example(plot)
```

```
example(boxplot)
```

```
example(hist)
```

```
example(persp)
```

Basic graph functions in R

Generic function ***plot*** and its methods:

plot

Generic X-Y Plotting

plot.data.frame

Plot Method for Data Frames

plot.default

The Default Scatterplot Function

plot.design

Plot Univariate Effects of a
'Design' or Model

plot.factor

Plotting Factor Variables

plot.formula

Formula Notation for Scatterplots

plot.histogram

Plot Histograms

plot.table

Plot Methods for 'table' Objects

plot.window

Set up World Coordinates for
Graphics Window

plot.xy

Basic Internal Plot Function

Basic graph functions in R

Axis	Generic function to add an Axis to a Plot
abline	Add Straight Lines to a Plot
arrows	Add Arrows to a Plot
assocplot	Association Plots
axTicks	Compute Axis Tickmark Locations
axis	Add an Axis to a Plot
axis.POSIXct	Date and Date-time Plotting Functions
barplot	Bar Plots
box	Draw a Box around a Plot
boxplot	Box Plots
bxp	Draw Box Plots from Summaries
cdplot	Conditional Density Plots
contour	Display Contours
coplot	Conditioning Plots
curve	Draw Function Plots
dotchart	Cleveland Dot Plots
filled.contour	Level (Contour) Plots
fourfoldplot	Fourfold Plots
frame	Create / Start a New Plot Frame
graphics-package	The R Graphics Package
grid	Add Grid to a Plot
hist	Histograms
hist.POSIXt	Histogram of a Date or Date-Time Object

Basic graph functions in R

identify

image

layout

legend

lines

locator

matplot

mosaicplot

mtext

pairs

panel.smooth

par

persp

pie

Identify Points in a Scatter Plot

Display a Color Image

Specifying Complex Plot Arrangements

Add Legends to Plots

Add Connected Line Segments to a Plot

Graphical Input

Plot Columns of Matrices

Mosaic Plots

Write Text into the Margins of a Plot

Scatterplot Matrices

Simple Panel Plot

Set or Query Graphical Parameters

Perspective Plots

Pie Charts

Basic graph functions in R

points

polygon

rect

rug

screen

segments

spineplot

stars

stem

stripchart

strwidth

sunflowerplot

symbols

text

title

xinch

Add Points to a Plot

Polygon Drawing

Draw One or More Rectangles

Add a Rug to a Plot

Creating and Controlling Multiple Screens on a Single Device

Add Line Segments to a Plot

Spine Plots and Spinograms

Star (Spider/Radar) Plots and Segment Diagrams

Stem-and-Leaf Plots

1-D Scatter Plots

Plotting Dimensions of Character Strings and Math Expressions

Produce a Sunflower Scatter Plot

Draw Symbols (Circles, Squares, Stars, Thermometers, Boxplots) on a Plot

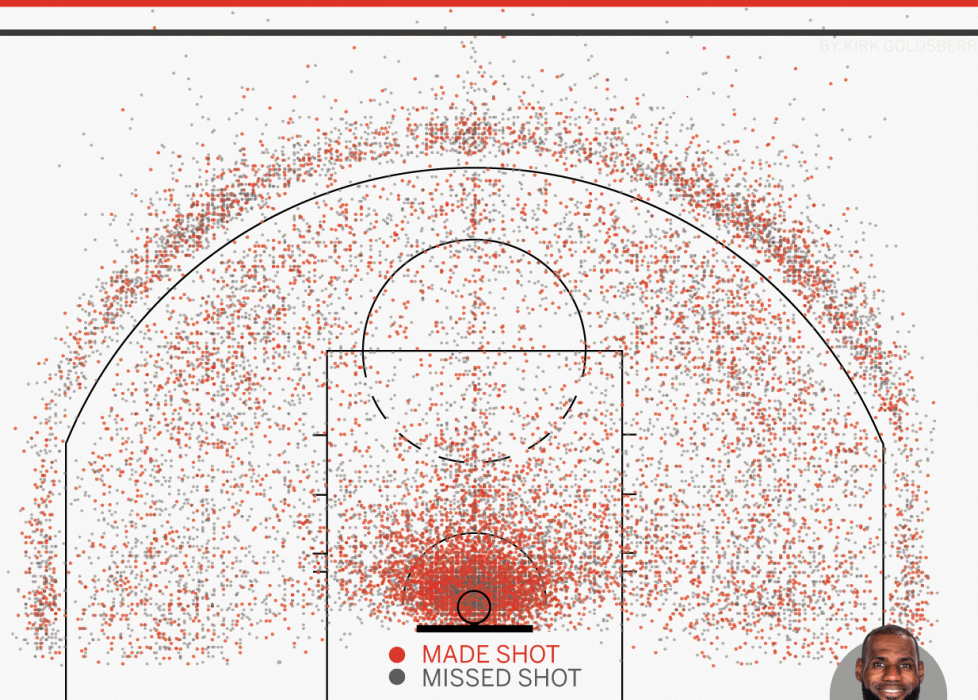
Add Text to a Plot

Plot Annotation

Graphical Units

The Atlas of LeBron James

CAREER FIELD GOAL ATTEMPTS



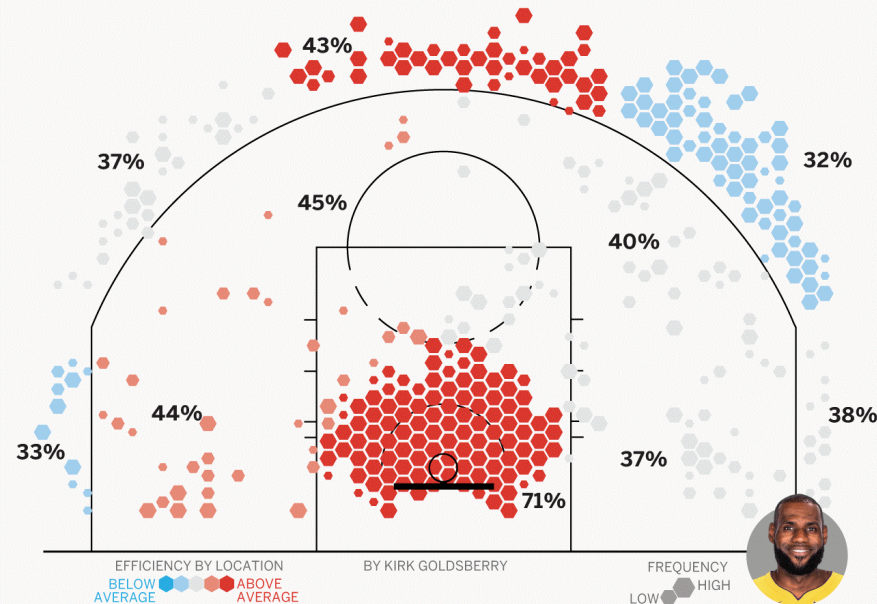
CAREER FIELD GOALS: 11,280 OF 22,383, 50.4%



STILL THE KING OF THE PAINT

Last season James led the NBA in scoring in the paint for the 6th time in the last 10 seasons

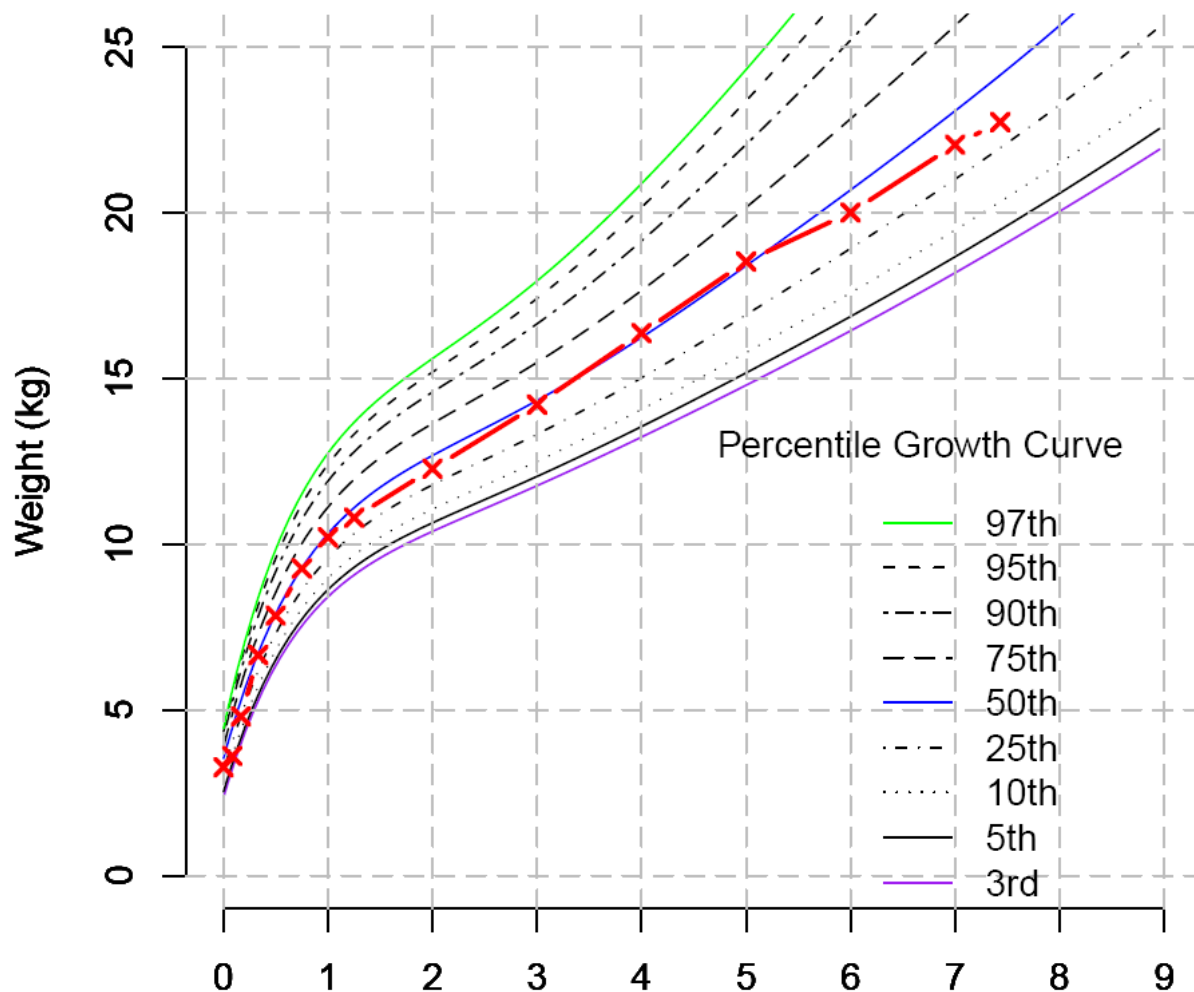
2017-18 SEASON



https://www.espn.com/nba/story/_/id/24999715/lebron-james-atlas-kirk-goldsberry-king-evolving-offensive-game

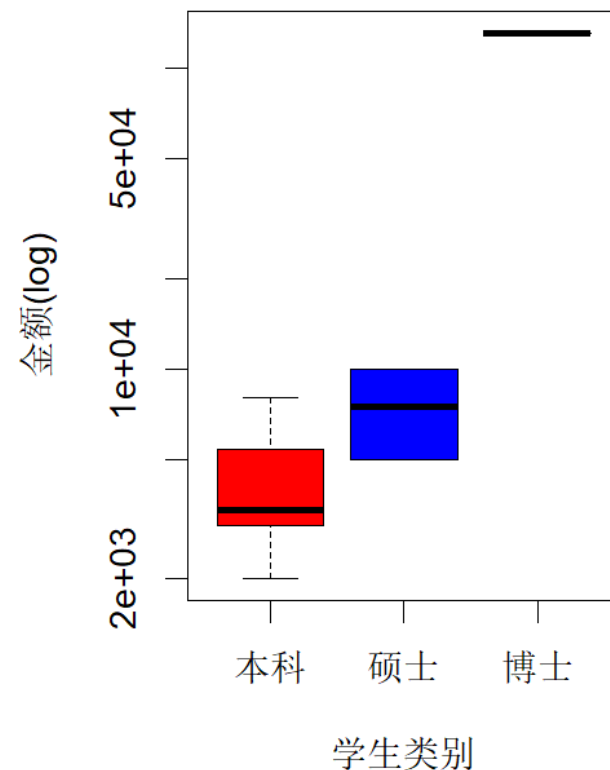
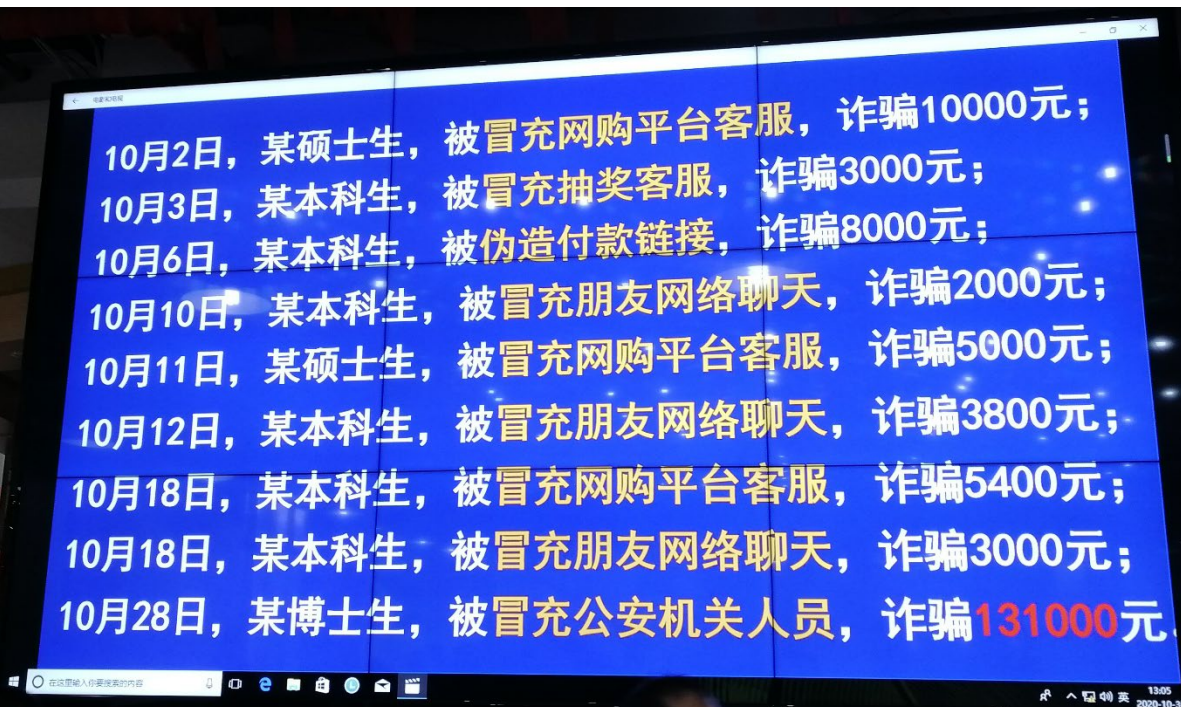
Florence Yong's son's weight

J Lo's Weight monitoring, on CDC Clinical Weight Growth Chart

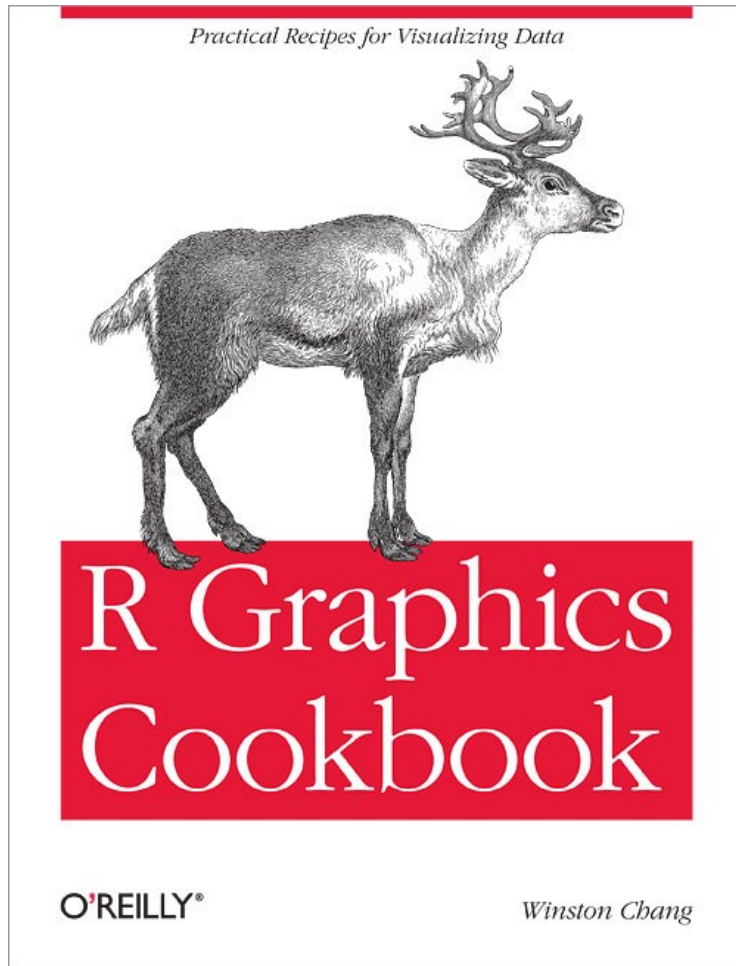


一位同学的例子：把R分析和作图应用到日常生活中

课后练习：用R绘图生活中的数据



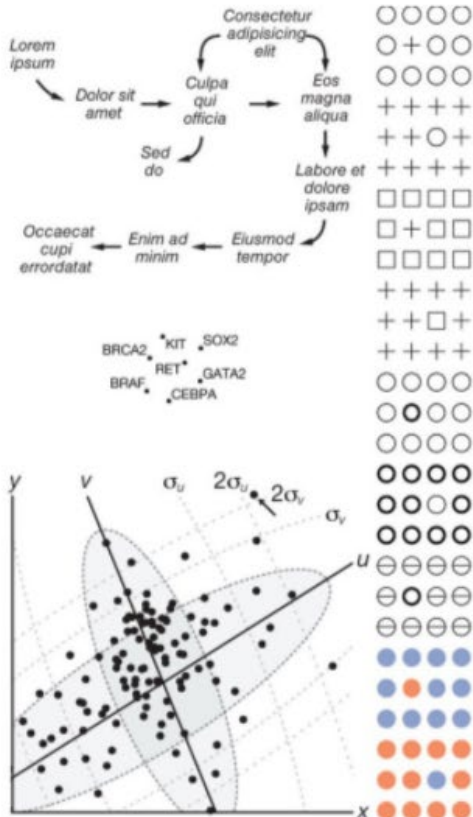
搜索和利用网上学习资源



- **R Graphics Cookbook**, a practical guide that provides more than **150 recipes to help you generate high-quality graphs quickly**, without having to comb through all the details of R's graphing systems.
- **Each recipe tackles a specific problem** with a solution you can apply to your own project, and includes a discussion of how and why the recipe works.

<https://r-graphics.org/>

《Nature Methods》做图和统计专栏



Elements of a figure

Typography – April 2011

Choose typefaces, sizes and spacing to clarify the structure and meaning of the text

Axes, ticks and grids – March 2013

Make navigational elements distinct and unobtrusive to maintain visual priority of data

Labels and callouts – April 2013

Figure labels require the same consistency and alignment in their layout as text

Plotting symbols – June 2013

Choose distinct symbols that overlap without ambiguity and communicate relationships in data

Arrows – September 2011

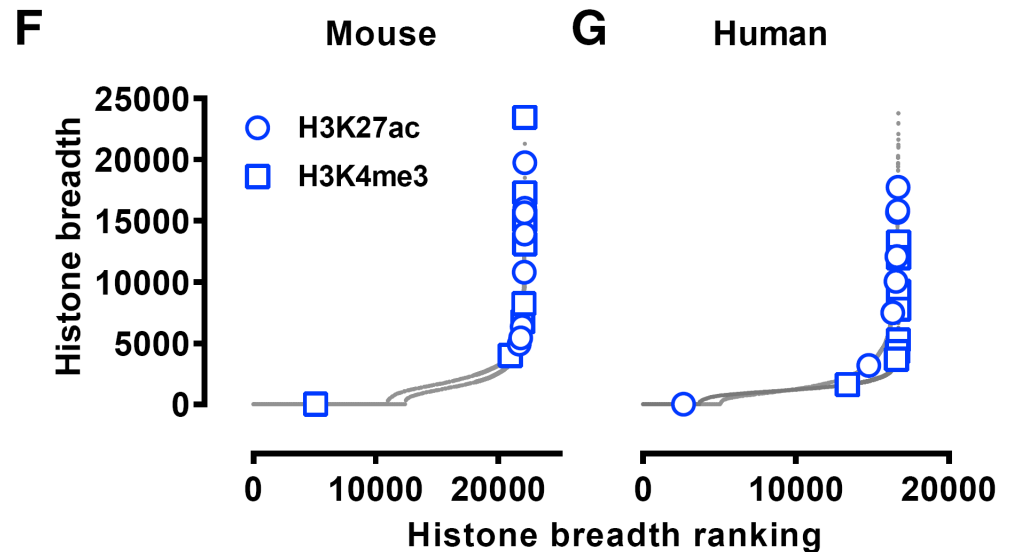
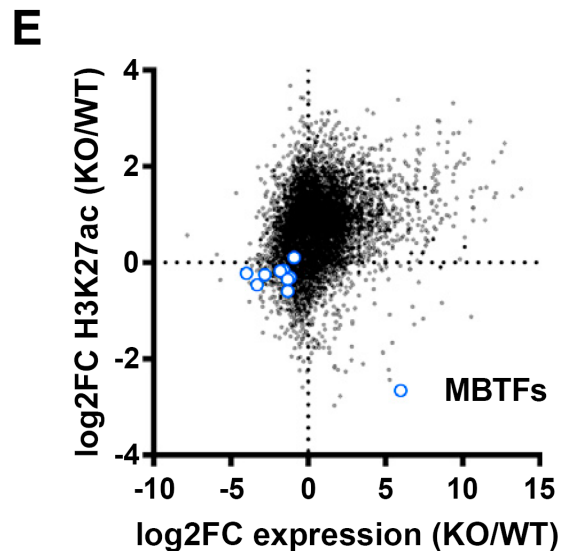
Use well-proportioned arrows sparingly and consistently as a guide through complex information

<http://blogs.nature.com/methagora/2013/07/data-visualization-points-of-view.html>
<https://www.nature.com/collections/qgghqm/pointsofsignificance>

分析法则 2：作图基本要素

- 标注X轴、Y轴的**数据含义和单位**
- 用图例注释点或符号的**形状和颜色**
- **突出显示**需要强调的部分

作图思路：读者不用读图注（甚至正文），就能理解你的结果和发现



课后练习

1. 阅读文献《Science 15 Variation in cancer risk among tissues can be explained by the number of stem cell divisions》、《Science 17 Stem cell divisions, somatic mutations, cancer etiology, and cancer prevention》（见课件“癌症运气”目录）
2. 思考上文作者的假设、数据分析和结论是否合理。相关学术讨论见：
<https://www.ncbi.nlm.nih.gov/pubmed/25554788> （“Comment in”部分）

课后练习

3. 数据分析练习

a) 《统计学习导论》，2.4节（习题），p.38，第8题

b) 结合课堂内容，运行R代码`regression.R`，理解所用函数的功能和输出结果。根据这些结果，课件中“Advertising营销问题”页面的哪一些问题可以回答？

c) 练习使用 R 基础作图函数（课件页面“Basic graph functions in R”），运行它们的帮助页面中的示例程序，理解结果。（如`?pie`，`example(pie)`）